

CHEMICAL EQUILIBRIUM

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THEORY

1. INTRODUCTION

Equilibrium represents the state of a process in which the properties like temperature, pressure, concentration of the system do not show any change with the passage of time.

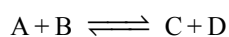
If the opposing processes involve only physical changes, the equilibrium is called physical equilibrium. If the opposing processes involve chemical changes, i.e., the opposing processes are chemical reactions, the equilibrium is called chemical equilibrium.

2. EQUILIBRIA IN CHEMICAL PROCESSES

2.1 Reversible Reactions

A reaction in which not only the reactants react to form the products under certain conditions but also the products react to form reactants under the same conditions is called a reversible reaction. In other words a reaction which takes place not only in the forward direction but also in the backward direction under the same conditions is called a reversible reaction.

A reversible reaction between A and B to form C and D is represented as :

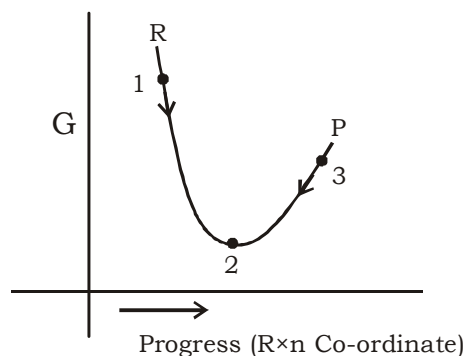


2.2 Irreversible Reactions

If a reaction cannot take place in the reverse direction, i.e., the products formed do not react to give back the reactants under the same conditions, it is called an irreversible reaction.



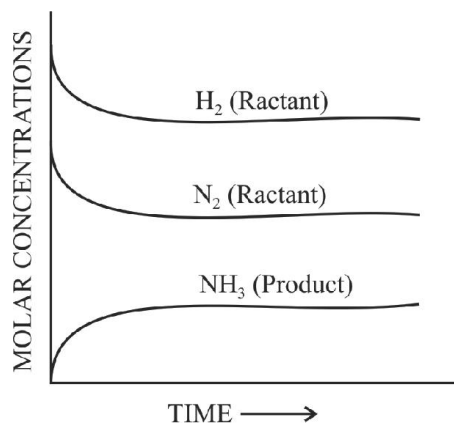
2.3 Concept of Chemical Equilibrium



- It is the state of minimum Gibb's energy
- $dG = 0$ and $\Delta G = 0$ at this state
- Rate of forward reaction = Rate of backward reaction
- This equilibrium is dynamic and stable in nature

Dynamic Nature of Chemical Equilibrium

In the Haber's process, starting with definite amounts of N_2 and H_2 and carrying out the reaction at a particular temperature, when equilibrium is attained, the concentrations of N_2 , H_2 and NH_3 become constant. If the experiment is repeated by taking deuterium (D_2) in place of H_2 but with the same amounts and exactly similar conditions as before, equilibrium is attained containing D_2 and ND_3 in place of H_2 and NH_3 but in the same amounts. Now, if the two reaction mixtures are mixed, then after some time, it is found that the concentrations of ammonia and hydrogen are same except that now all forms of ammonia (i.e., NH_3 , NH_2D , NHD_2 , ND_3) and all forms of hydrogen (i.e., H_2 , HD , D_2) are present. This shows that at equilibrium, the reaction is still going on, i.e., equilibrium is dynamic in nature.



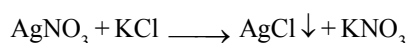
Representing the attainment of equilibrium for the reaction
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

2.4 Characteristics of Chemical Equilibrium

- At equilibrium, the concentration of each of the reactants and the products becomes constant.
- At equilibrium, the rate of forward reaction becomes equal to the rate of backward reaction and hence the equilibrium is dynamic in nature.
- A chemical equilibrium can be established only if none of the products is allowed to escape out or separate out as a solid.

e.g. if CO_2 gas escapes out in case of decomposition of CaCO_3 , the reaction will no longer remain reversible.

Similarly, the reaction is irreversible if one of the products separates out as solid,



- Chemical equilibrium can be attained from either direction, i.e., from the direction of the reactants as well as from the direction of the products.

3. RATE OF A REACTION

Average Rate = Change in concentration/Time taken
 $= \Delta c / \Delta t$

Δc = Final Concentration – Initial concentration

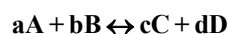
3.1 Instantaneous rate

Lim $\Delta c / \Delta t = dc / dt$

$\Delta t \rightarrow 0$

Units of rate : conc/time or mol/Ls

3.2 Overall rate of a reaction



Overall Rate =

$$-(1/a) \Delta \text{A} / \Delta t = -(1/b) \Delta \text{B} / \Delta t = +(1/c) \Delta \text{C} / \Delta t = +(1/d) \Delta \text{D} / \Delta t$$

Overall rate :

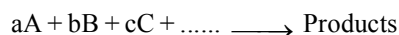
Rate of forward reaction – Rate of backward reaction

“At equilibrium the overall rate of a reversible reaction becomes zero”.

4. EQUILIBRIUM CONSTANT (K)

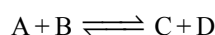
4.1 Law of Mass Action

The rate of a reaction is proportional to the product of the active masses of the reactants, each raised to the power equal to its stoichiometric coefficient as represented by the balanced chemical equation.



$$\text{Rate of reaction} \propto [\text{A}]^a [\text{B}]^b [\text{C}]^c \dots$$

Law of Chemical Equilibrium is a result obtained by applying the Law of Mass Action to a reversible reaction in equilibrium.



$$\text{Rate of the forward reaction} \propto [\text{A}] [\text{B}] = k_f [\text{A}] [\text{B}]$$

$$\text{Rate of the backward reaction} \propto [\text{C}] [\text{D}] = k_b [\text{C}] [\text{D}]$$

At equilibrium,

$$\text{Rate of forward reaction} = \text{Rate of backward reaction}$$

$$k_f [\text{A}] [\text{B}] = k_b [\text{C}] [\text{D}] \text{ or } \frac{[\text{C}] [\text{D}]}{[\text{A}] [\text{B}]} = \frac{k_f}{k_b} = K$$

At constant temperature, as k_f and k_b are constant, therefore,

$\frac{k_f}{k_b} = K$ is also constant at constant temperature and is called ‘Equilibrium constant’.

The product of the molar concentrations of the products, each raised to the power equal to its stoichiometric coefficient divided by the product of the molar concentrations of the reactants, each raised to the power equal to its stoichiometric coefficient is constant at constant temperature and is called Equilibrium constant.

4.2 Characteristics of Equilibrium Constant

- (i) The value of the equilibrium constant for a particular reaction is always constant depending only upon the temperature of the reaction and is independent of the concentrations of the reactants with which we start or the direction from which the equilibrium is approached.
- (ii) If the reaction is reversed, the value of the equilibrium constant is inverted.
- (iii) If the equation (having equilibrium constant K) is divided by 2, the equilibrium constant for the new equation is the square root of K (i.e., $\sqrt{K}$).
- (iv) If the equation (having equilibrium constant K) is multiplied by 2, the equilibrium constant for the new equation is the square of K (i.e., K^2).
- (v) If the equation (having equilibrium constant K) is written in two steps (having equilibrium constant K_1 and K_2) then $K_1 \times K_2 = K$.
- (vi) The value of the equilibrium constant is not affected by the addition of a catalyst to the reaction.

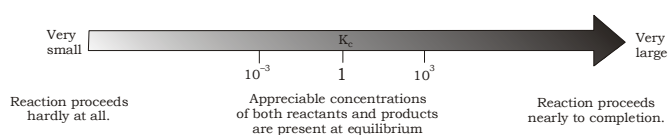
This is because the catalyst increases the speed of the forward reaction and the backward reaction to the same extent.

4.3 Extent of Reaction

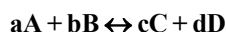
High value of K_c ($K_c > 10^3$) $\rightarrow$ At equilibrium reaction is forward dominant

Low value of K_c ($K_c < 10^{-3}$) $\rightarrow$ At equilibrium reaction is backward dominant

Moderate value of K_c (between 10^3 and 10^{-3}) $\rightarrow$ At equilibrium neither direction dominates



4.4 Reaction Quotient (Q)



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$Q > K_c$: Reaction will tend towards backward direction

$Q < K_c$: Reaction will tend towards forward direction

$Q = K_c$: Reaction will be at equilibrium

Note :

Q is a variable which always approaches K_{eq} which is a constant.

4.5 Calculating Equilibrium Concentrations

Suppose we are given the following data :

1. The balanced reaction and value of K_c
2. The initial concentration of the reactants, or the initial moles
3. Volume of the container

And we need to find the final equilibrium concentration of the reactants and products. Then we can follow the following steps :

Step-1

Write down the balanced chemical equation for the reaction

Step-2

Under every reactant and product, write down the initial moles/concentration

Step-3

Subtract the amount reacting and add the amount produced in terms of a variable x and note down the equilibrium concentration in terms of x . If we are dealing in moles then we will need to divide the moles by volume to obtain concentrations.

Step-4

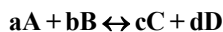
Substitute the equilibrium concentration in the expression of K_c and equate it to the value of K_c .

Step-5

Solve the above equation and calculate the value of x and in case of multiple solutions select the value which is sensible from reaction point of view. Then back substitute

the value of x in the equilibrium concentration expression and obtain the actual value of the same.

4.6 Equilibrium constant K_p



$$K_p = (P_C^c \times P_D^d) / (P_A^a \times P_B^b)$$

where P_C , P_D , P_A , P_B are partial pressures of A, B, C and D respectively.

$$K_p = K_c (RT)^{\Delta n}$$

$$R = 0.0821 \text{ L atm/mol-K}$$

4.7 Relationship of K_{eq} and ΔG

For any reaction :

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \text{Where } Q \text{ is the reaction quotient.}$$

At equilibrium $\Delta G = 0$

$$\Delta G^\circ = -RT \ln K_{eq} = -2.303RT \log_{10} K_{eq}$$

where K_{eq} is generally taken as K_p . It depends on the definition of standard values to define ΔG° .

If the standard active masses are taken as 1 M each then we will take $K_{eq} = K_c$ and if they are taken as 1 atm each then we will take $K_{eq} = K_p$.

4.8 Dependence of K_{eq} on temperature

Van't Hoff Equation :

$$\ln (K_2/K_1) = (\Delta H/R) (1/T_1 - 1/T_2) \quad \text{or}$$

$$\log_{10} (K_2/K_1) = (\Delta H/2.303R) (1/T_1 - 1/T_2)$$

Exothermic Reactions

$$\Delta H = \text{negative}$$

On increasing the temperature the equilibrium constant will decrease i.e. The reaction will become more backward dominant.

Endothermic Reaction

$$\Delta H = \text{positive}$$

On increasing the temperature the equilibrium constant will increase. The reaction will become more forward dominant.

POINTS TO REMEMBER

Whether the reaction is exothermic or endothermic on increasing the temperature, it will tend towards endothermic direction.

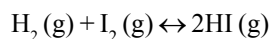
5. HOMOGENOUS EQUILIBRIA

Reactions in which all reactants and products are in the same phase Homogenous Reactions can further be divided into three sub-categories :

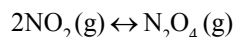
5.1 Those reactions where gaseous moles increase ($\Delta n = +ve$)



5.2 Those reactions where gaseous moles remain the same ($\Delta n = 0$)



5.3 Those reactions where gaseous moles decrease ($\Delta n = -ve$)



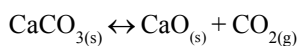
6. DEGREE OF DISSOCIATION, α

It is defined as the fraction of molecules dissociating. For example, if 100 molecules are present and only 40 dissociate then the degree of dissociation is 0.4 or 40%.

7. HETEROGENEOUS EQUILIBRIA

Category of reactions where various phases/states exist in the same reaction.

In presence of gases the activity of solids and excess liquids is constant. Therefore we can assume the active masses of solids and excess liquids as constant.



$$K_c = [CO_2]$$

$$K_p = P_{CO_2}$$

8. LE CHATELIER'S PRINCIPLE

If a disturbance is introduced in an equilibrium mixture it will behave so as to undo the disturbance and re-establish equilibrium.

8.1 Effect of Change of Concentration

If in a reaction in equilibrium, the concentration of any reactant is increased, the equilibrium shifts in the forward direction. On the other hand, if the concentration of any product is increased, the equilibrium shifts in the backward direction. The reverse happens if the concentrations are decreased.

8.2 Effect of change of temperature

Exothermic reactions are favoured by low temperature whereas endothermic reactions are favoured by high temperature.

8.3 Effect of change of pressure

Low pressure favours those reactions which are accompanied by increase in total number of moles and high pressure favours those reactions which take place with decrease in total number of moles. However, pressure has no effect on an equilibrium reaction which proceeds with no change in total number of moles.

8.4 Effect of change in volume

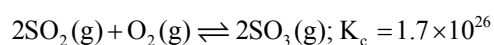
The effect of decrease of volume is equivalent to the effect of increase of pressure.

hence the effect of decrease in volume will be to shift the equilibrium in the direction in which the number of moles decreases.

8.5 Effect of a Catalyst

A catalyst increases the rate of the chemical reaction by making available a new low energy pathway for the conversion of reactants to products. It increases the rate of forward and reverse reactions that pass through the same transition state and does not affect equilibrium. Catalyst lowers the activation energy for the forward and reverse reactions by exactly the same amount. Catalyst does not affect the equilibrium composition of a reaction mixture. It does not appear in the balanced chemical equation or in the equilibrium constant expression.

eg contact process,



Practically the oxidation of SO_2 to SO_3 is very slow. Thus, platinum or divanadium penta-oxide (V_2O_5) is used as catalyst to increase the rate of the reaction.

8.6 Effect of adding an inert gas to a reaction mixture in equilibrium.

- (a) If the reaction takes place at constant volume addition of an inert gas will not change the molar concentrations of the reactants and products. Hence, the state of equilibrium will remain unaffected.
- (b) If the reaction takes place at constant pressure, addition of inert gas must accompany an increase in the total volume to keep pressure constant. Thus, reaction shifts towards larger number of moles.

SOLVED EXAMPLES

Example - 1

The following reaction has attained equilibrium



What will happen if (i) volume of the reaction vessel is suddenly reduced to half? (ii) the partial pressure of hydrogen is suddenly doubled? (iii) an inert gas is added to the system?

Sol. $K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}, K_p = \frac{P_{\text{CH}_3\text{OH}}}{P_{\text{CO}} \times p_{\text{H}_2}^2}$

- (i) When volume of the vessel is reduced to half, the concentration of each reactant or product becomes double.

Thus, $Q_c = \frac{2[\text{CH}_3\text{OH}]}{2[\text{CO}](2[\text{H}_2])^2} = \frac{1}{4}K_c$. As $Q_c < K_c$, equilibrium will shift in the forward direction, producing more of CH_3OH to make $Q_c = K_c$.

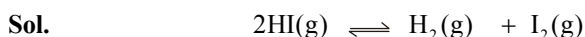
$$(ii) \quad Q_p = \frac{P_{CH_3OH}}{p_{CO}} \times (2p_{H_2})^2 = \frac{1}{4} K_p.$$

Again $Q_p < K_p$, equilibrium will shift in the forward direction to make $Q_p = K_p$.

- (iii) As volume remains constant, molar concentration will not change. Hence there is no effect on the state of equilibrium.

Example - 2

A sample of HI (g) is placed in a flask at a pressure of 0.2 atm. At equilibrium, the partial pressure of HI (g) is 0.04 atm. What is K_p for the given equilibrium?



Initial pressure	0.2 atm	0	0
At eqm.	0.04 atm	$\frac{0.16}{2}$ atm	$\frac{0.16}{2}$ atm
		= 0.08 atm	

(Decrease in the pressure of HI = $0.2 - 0.04 = 0.16$ atm.)

$$\therefore K_p = \frac{p_{H_2} \times p_{I_2}}{p_{HI}^2} = \frac{0.08 \text{ atm} \times 0.08 \text{ atm}}{(0.04 \text{ atm})^2} = 4.0.$$

Example - 3

One mole of H_2O and one mole of CO are taken in a 10 litre vessel and heated at 725 K. At equilibrium 40% of water (by mass) reacts with CO according to the equation:



Calculate the equilibrium constant for the reaction.

Sol. At equilibrium $[\text{H}_2\text{O}] = \frac{1-0.40}{10} \text{ mol L}^{-1} = 0.06 \text{ mol L}^{-1}$,
 $[\text{CO}] = 0.06 \text{ mol L}^{-1}$.

$$[\text{H}_2] = \frac{0.4}{10} \text{ mol L}^{-1} = 0.04 \text{ mol L}^{-1},$$

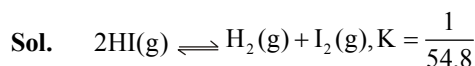
$$[\text{CO}_2] = 0.04 \text{ mol L}^{-1}$$

$$K = \frac{[H_2][CO_2]}{[H_2O][CO]} = \frac{0.04 \times 0.04}{0.06 \times 0.06} = 0.444$$

Example - 4

At 700 K, equilibrium constant for the reaction

$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ is 54.8. If 0.5 mol L^{-1} of $\text{HI}(\text{g})$ is present at equilibrium at 700K, what are the concentrations of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ assuming that we initially started with $\text{HI}(\text{g})$ and allowed it to reach equilibrium at 700 K.



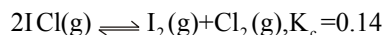
At equilibrium $[\text{HI}] = 0.5 \text{ mol L}^{-1}$,

$$[\text{H}_2] = [\text{I}_2] = x \text{ mol L}^{-1} \therefore K = \frac{x \times x}{(0.5)^2} = \frac{1}{54.8}$$

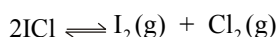
This gives $x = 0.068$. i.e., $[H_2] = [I_2] = 0.068 \text{ mol L}^{-1}$

Example - 5

What is the equilibrium concentration of each of substances in the equilibrium when the initial concentration of ICl was 0.78 M?



Sol. Suppose at equilibrium, $[\text{I}_2] = [\text{Cl}_2] = x \text{ mol L}^{-1}$. Then



Initial conc. 0.78M 0 0

At eqm. 0.78-2x x x

$$K_c = \frac{[\text{I}_2][\text{Cl}_2]}{[\text{ICl}]^2} \therefore 0.14 = \frac{x \times x}{(0.78 - 2x)^2}$$

$$\text{or } x^2 = 0.14(0.78 - 2x)^2 \text{ or } \frac{x}{0.78 - 2x} = \sqrt{0.14} = 0.374$$

$$\text{or } x = 0.292 - 0.748x \text{ or } 1.748x = 0.292 \text{ or } x = 0.167$$

Hence, at equilibrium,

$$[\text{I}_2] = [\text{Cl}_2] = 0.167\text{M}, [\text{ICl}] = 0.78 - 2 \times 0.167\text{M} = 0.446\text{M}$$

Example - 6

Ethyl acetate is formed by the reaction between ethanol and acetic acid and the equilibrium represented as:

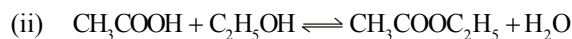


(i) Write the concentration ratio (reaction quotient), Q_c for this reaction (Note: water is not in excess and is not a solvent in this reaction)

(ii) At 293 K, if one starts with 1.00 mol of acetic acid and 0.18 mol of ethanol, there is 0.171 mol of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant.

(iii) Starting with 0.5 mol of ethanol and 1.0 mol of acetic acid and maintaining it at 293 K, 0.214 mol of ethyl acetate is found after some time. Has equilibrium been reached?

$$\text{Sol. (i) } Q_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]}$$



Initial 1.00 mol 0.180 mol

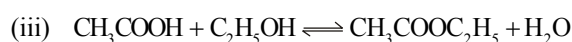
At eqm. 1-0.171 0.180-0.171 0.171 mol 0.171 mol

$$= 0.829 \text{ mol} = 0.009 \text{ mol}$$

Molar

concs 8.29/V 0.009/V 0.171/V 0.171/V

$$K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]} = \frac{(0.171/V)(0.171/V)}{(0.829/V)(0.009/V)} = 3.92$$



Initial 1.000 mol 0.500 mol

After time t 1-0.214 0.500-0.214 0.214 mol 0.214 mol

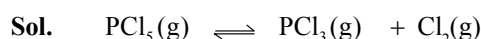
$$= 0.786 \text{ mol} = 0.286 \text{ mol}$$

$$\text{Reaction quotient } (Q_c) = \frac{(0.214/V)(0.214/V)}{(0.786/V)(0.286/V)} = 0.204$$

As $Q_c \neq K_c$, equilibrium has not been attained

Example - 7

A sample of pure PCl_5 was introduced into an evacuated vessel at 473 K. After equilibrium was attained, concentration of PCl_5 was found to be $0.5 \times 10^{-1} \text{ mol L}^{-1}$. If value of K_c is 8.3×10^{-3} , what are the concentrations of PCl_3 and Cl_2 at equilibrium?



At eqm $0.5 \times 10^{-1} \text{ mol L}^{-1}$ x mol L^{-1} x mol L^{-1} (suppose)

$$\therefore K_c = \frac{x^2}{0.5 \times 10^{-1}} = 8.3 \times 10^{-3} \text{ (Given)}$$

$$\text{or } x^2 = (8.3 \times 10^{-3})(0.5 \times 10^{-1}) = 4.15 \times 10^{-4} \text{ or}$$

$$x = \sqrt{4.15 \times 10^{-4}} = 2.04 \times 10^{-2} \text{ M} = 0.02 \text{ M}$$

$$\text{Hence, } [\text{PCl}_3]_{\text{eq}} = [\text{Cl}_2]_{\text{eq}} = 0.02 \text{ M}$$

Example - 8

Equilibrium constant, K_c for the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at 500 K is 0.061. At a particular time, the analysis shows that composition of the reaction mixture is $3.0 \text{ mol L}^{-1} N_2$, $2.0 \text{ mol L}^{-1} H_2$ and $0.5 \text{ mol L}^{-1} NH_3$. Is the reaction at equilibrium? If not, in which direction does the reaction tend to proceed to reach equilibrium?

Sol. $Q_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.5)^2}{(3.0)(2.0)^3} = 0.0104$

As $Q_c \neq K_c$, reaction is **not in equilibrium**

As $Q_c < K_c$, reaction will proceed in the **forward direction**.

Example - 9

The equilibrium constant for the reaction $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$ at 1024 K is 1.6×10^5 . Find the equilibrium pressure of all gases if 10.0 bar of HBr is introduced into a sealed container at 1024 K.

Sol. $2HBr(g) \rightleftharpoons H_2(g) + Br_2(g)$. $K = \frac{1}{(1.6 \times 10^5)}$

Initial 10 bar

At eqm. $10-p$ $p/2$ $p/2$

$$K_p = \frac{(p/2)(p/2)}{(10-p)^2} = \frac{1}{1.6 \times 10^5} \Rightarrow \frac{p^2}{4(10-p)^2} = \frac{1}{1.6 \times 10^5}$$

Taking square root of both sides, we get

$$\frac{p}{2(10-p)} = \frac{1}{4 \times 10^2} \text{ or } 4 \times 10^2 p = 2(10-p) \text{ or}$$

$$402p = 20 \text{ or } p = \frac{20}{402} = 4.98 \times 10^{-2} \text{ bar}$$

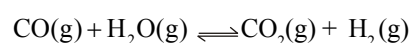
Hence, at equilibrium $p_{H_2} = p_{Br_2}$

$$= p/2 = 2.5 \times 10^{-2} \text{ bar, } p_{HBr} = 10 - p \approx 10 \text{ bar}$$

Example - 10

Dihydrogen gas used in Haber's process is produced by reacting methane from natural gas with high temperature steam. The first stage of two stage reaction involves the formation of CO and H_2 . In second stage, CO formed in first state is reacted with more steam in water gas shift reaction, $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$. If a reaction vessel at 400°C is charged with an equimolar mixture of CO and steam such that $p_{CO} = p_{H_2O} = 4.0 \text{ bar}$, what will be the partial pressure of H_2 at equilibrium? $K_p = 0.1$ at 400°C.

Sol. Suppose the partial pressure of H_2 at equilibrium = p bar



Initial pressure 4.0 bar 4.0 bar

At eqm $(4-p)$ $(4-p)$ p p

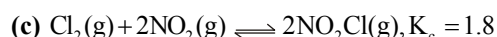
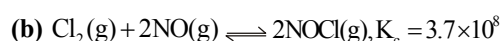
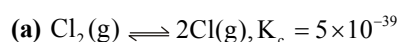
$$K_p = \frac{p^2}{(4-p)^2} = 0.1 (\text{given}) \therefore \frac{p}{4-p} = \sqrt{0.1} = 0.316$$

$p = 1.264 - 0.316p$ or $1.316p = 1.264$ or $p = 0.96 \text{ bar}$. Hence,

$$(p_{H_2})_{eq} = 0.96 \text{ bar}$$

Example - 11

Predict which of the following reaction will have appreciable concentration of reactants and products:



Sol. For reaction (c), as K_c is neither high nor very low, reactants and products will be present in comparable amounts.

Example - 12

The value of K_c for reaction, $3O_2(g) \rightleftharpoons 2O_3(g)$, is 2.0×10^{-50} at 25°C. If the equilibrium concentration of O_2 in air at 25°C is 1.6×10^{-2} , what is the concentration of O_3 ?

Sol. $K_c = \frac{[O_3]^2}{[O_2]^3} \therefore 2.0 \times 10^{-50} = \frac{[O_3]^2}{(1.6 \times 10^{-2})^3}$

$$\text{or } [\text{O}_3]^2 = (2.0 \times 10^{-50}) (1.6 \times 10^{-2})^3 = 8.192 \times 10^{-56} \text{ or}$$

$$[\text{O}_3] = 2.86 \times 10^{-28} \text{ M}$$

Example - 13

The reaction, $\text{CO(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_4\text{(g)} + \text{H}_2\text{O(g)}$, is at equilibrium at 1300 K in a 1 L flask. It also contains 0.30 mol of CO, 0.10 mol of H_2 and 0.02 mol of H_2O and an unknown amount of CH_4 in the flask. Determine the concentration of CH_4 in the mixture. The equilibrium constant, K_c , for the reaction at the given temperature is 3.90.

$$\text{Sol. } K_c = \frac{[\text{CH}_4][\text{H}_2\text{O}]}{[\text{CO}][\text{H}_2]^3}$$

$$\therefore 3.90 = \frac{[\text{CH}_4](0.02)}{(0.30)(0.10)^3} \quad (\text{molar conc} = \text{No. of moles})$$

because volume of flask = 1L)

$$\text{or } [\text{CH}_4] = 0.0585 \text{ M} = 5.85 \times 10^{-2} \text{ M}$$

Example - 14

At emperature T, a compound $\text{AB}_2\text{(g)}$ dissociates according to the reaction



with a degree of dissociation, x , which is small compared with unity. Deduce the expression for x in terms of the equilibrium constant K_p and the total pressure, P .

$$\text{Sol. } 2\text{AB}_2 \rightleftharpoons 2\text{AB(g)} + \text{B}_2\text{(g)}$$

Initial	1 mole	0	0
At equilibrium	$1-x$	$2x$	x

Total number of moles at equilibrium = $1-x+2x+x=1+2x$

$$p_{\text{AB}_2} = \frac{1-x}{1+2x} \times P, p_{\text{AB}} = \frac{2x}{1+2x} \times P, p_{\text{B}_2} = \frac{x}{1+2x} \times P$$

$$\therefore K_p = \frac{p_{\text{AB}}^2 \times p_{\text{B}_2}}{p_{\text{AB}_2}^2} = \left(\frac{2x \times P}{1+2x} \right)^2 \times \left(\frac{x \times P}{1+2x} \right) \left(\frac{1-x}{1+2x} \times P \right)^2$$

$$\text{or } K_p = \frac{4x^3 P}{(1-x)(1+2x)}$$

Neglection x in comparison to unity,

$$K_p = 4x^3 P \text{ or } x = \left(\frac{K_p}{4P} \right)^{1/3}$$

Example - 15

Prove that the pressure necessary to obtain 50% dissociation of PCl_5 at 500 K is numerically equal to three times the value of the equilibrium constant, K_p .

$$\text{Sol. } \text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$$

Initial moles	1	0	0
Moles at eqm.	$1-0.5=0.5$	0.5	0.5

Total = 1.5 moles

If P is the total required pressure, then

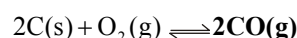
$$p_{\text{PCl}_5} = \frac{0.5}{1.5} \times P = \frac{P}{3}, p_{\text{PCl}_3} = \frac{0.5}{1.5} \times P = \frac{P}{3}$$

$$p_{\text{Cl}_2} = \frac{0.5}{1.5} \times P = \frac{P}{3}$$

$$\therefore K_p = \frac{p_{\text{PCl}_3} \times p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{(P/3)(P/3)}{(P/3)} = \frac{P}{3} \text{ or } P = 3K_p$$

Example - 16

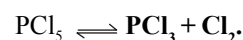
What is the effect of reducing the volume on the system described below?



Sol. On reducing the volume, the pressure will increase. By Le Chatelier's principle, equilibrium will shift to the side accompanied by decrease of pressure, i.e., decrease in the number of gaseous moles, i.e., backward direction.

Example - 17

The degree of dissociation is 0.4 at 400 K and 1.0 atm for the gaseous reaction



Assuming ideal behaviour of all the gases, calculate the density of the equilibrium mixture at 400 K and 1.0 atmosphere. (Atomic mass of P = 31.0 and Cl = 35.5)

$$\text{Sol. } \text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$$

Initial	1 mole	0	0
At eqm.	0.6	0.4	0.4

Average molecular mass of mixture

$$= \frac{0.6(31+5 \times 35.5) + 0.4(31+3 \times 35.5) + 0.4(2 \times 35.5)}{0.6+0.4+0.4}$$

$$= \frac{125.1+55+42.6}{1.4} = 148.92$$

For ideal gases,

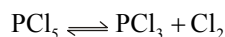
$$PV = nRT = \frac{W}{M}RT \text{ or } PM = \frac{W}{V}RT = dRT$$

$$\therefore d = \frac{PM}{RT} = \frac{1 \times 148.92}{0.08205 \times 400} = 4.5374 \text{ g L}^{-1}$$

Example - 18

At 540 K, 0.10 mol of PCl_5 are heated in a 8.0 L flask. The pressure of the equilibrium mixture is found to be 1.0 atm. Calculate K_p and K_c for the reaction.

Sol.



Initial 0.1 mol

At eqm. 0.1 - x x x

Total no. of moles at eqm. = 0.1 + x

$$PV = nRT, \text{ i.e., } n = \frac{PV}{RT} = \frac{1 \times 8}{0.0821 \times 540} = 0.18$$

$$\therefore 0.1 + x = 0.18 \text{ or } x = 0.08$$

$$\therefore K_c = \frac{(0.08/8)(0.08/8)}{(0.02/8)} = 0.04$$

$$K_p = K_c(RT)^{\Delta n} = 0.04(0.0821 \times 540)^1 = 1.77$$

Example - 19

An equilibrium mixture,

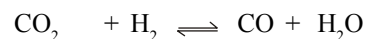
$\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$, present in a vessel of one litre capacity at 1000 K was found to contain 0.4 mole of CO , 0.3 mole of H_2O , 0.2 mole of CO_2 and 0.6 mole of H_2 . If it is desired to increase the concentration of CO to 0.6 mole by adding CO_2 into the vessel, how many moles of it must be added into equilibrium mixture at constant temperature in order to get this change.

Sol. Step 1. To calculate K_c of the reaction.

$$K_c = \frac{[\text{CO(g)}][\text{H}_2\text{(g)}]}{[\text{CO}_2\text{(g)}][\text{H}_2\text{O(g)}]} = \frac{0.2 \times 0.6}{0.4 \times 0.3} = 1$$

Step 2. To calculate extra CO_2 to be added

Suppose extra CO_2 to be added = x mole. Then writing the reverse reaction, we have



initial moles after

addition 0.2 + x 0.6 0.4 0.3

Moles (Molar

Conc.) at new (0.2+x-0.2) (0.6-0.2) 0.6 (0.3+0.2)

equilibrium = x = 0.4 = 0.5

(V=1L) (given)

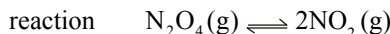
$$K_c' = \frac{1}{K_c} = \frac{0.6 \times 0.5}{x \times 0.4} = 1 \text{ or } x = 0.75 \text{ moles}$$

Example - 20

The degree of dissociation of N_2O_4 into NO_2 at one atmospheric pressure and 313 K is 0.310. Calculate K_p of the dissociation reaction at this temperature. What will be the degree of dissociation at 10 atmospheric pressure at the same temperature?

Sol.

For the dissociation



Initial moles 1 0

Moles at eqm. 1-0.310=0.69 2×0.310=0.62 ($\alpha=0.310$, Given)

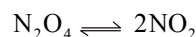
Total = 0.69+0.62=1.31 moles

At one atmospheric

pressure, $p_{\text{N}_2\text{O}_4} = \frac{0.69}{1.31} \times 1 \text{ atm}$, $p_{\text{NO}_2} = \frac{0.62}{1.31} \times 1 \text{ atm}$

$$K_p = \frac{p_{\text{NO}_2}^2}{p_{\text{N}_2\text{O}_4}} = \frac{(0.62/1.31 \text{ atm})^2}{(0.69/1.31 \text{ atm})} = 0.425 \text{ atm}$$

At 10 atmospheric pressure, suppose degree of dissociation = α .
Then



Initial moles 1 0

Moles at eqm 1- α 2 α , Total = 1+ α

$$p_{\text{N}_2\text{O}_4} = \frac{1-\alpha}{1+\alpha} \times 10\text{atm}, p_{\text{NO}_2} = \frac{2\alpha}{1+\alpha} \times 10\text{atm}$$

$$K_p = \frac{[20\alpha/(1+\alpha)]^2}{[10(1-\alpha)/(1+\alpha)]} = \frac{40\alpha^2}{1-\alpha(1+\alpha)} = \frac{40\alpha^2}{1-\alpha^2}$$

$$\therefore \frac{40\alpha^2}{1-\alpha^2} = 0.425$$

Neglecting α^2 in comparison to 1, $40\alpha^2 = 0.425$

or $\alpha = 0.103 = 10.3\%$

Example - 21

When α - D glucose is dissolved in water, it undergoes mutarotation to form an equilibrium mixture of α - D glucose and β - D glucose containing 63.6% of the latter. Calculate K_c for the mutarotation.

Sol. α -D glucose $\rightleftharpoons$ β - D glucose

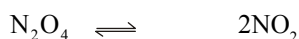
At equilibrium 36.4% 63.6%

$$K_c = \frac{63.6}{36.4} = 1.747$$

Example - 22

At 77°C and one atmospheric pressure, N_2O_4 is 70% dissociated into NO_2 . What will be the volume occupied by the mixture under these conditions if we start with 10g of N_2O_4 ?

Sol. Molar mass of $\text{N}_2\text{O}_4 = 28 + 64 = 92\text{g mol}^{-1}$



Initial moles $\frac{10}{92}$ 0

After dissociation $\frac{10}{92} - \frac{70}{100} \times \frac{10}{92}$ 2×0.076

$$= 0.109 - 0.076 = 0.033 = 0.152$$

Example - 23

0.1 mole of N_2O_4 (g) was sealed in a tube under atmospheric condition at 25°C. Calculate the number of moles of NO_2 (g) present if the equilibrium N_2O_4 (g) is reached after some time ($K_p = 0.14$).

Sol. N_2O_4 (g) $\rightleftharpoons$ 2NO_2 (g)

Initial amount 0.1 mole 0

At equilibrium $(0.1-x)$ $2x$, Total = $0.1+x$ moles

As $P=1$ atm, $p_{\text{N}_2\text{O}_4} = \frac{0.1-x}{0.1+x}$, $p_{\text{NO}_2} = \frac{2x}{0.1+x}$ atm

$$K_p = \frac{p_{\text{NO}_2}^2}{p_{\text{N}_2\text{O}_4}} = \frac{(2x/(0.1+x))^2}{(0.1-x)/(0.1+x)}$$

$$= \frac{4x^2}{(0.1-x)(0.1+x)} = \frac{4x^2}{0.01-x^2}$$

$$\therefore \frac{4x^2}{0.01-x^2} = 0.14 \text{ or } 4.14x^2 = 0.0014 \text{ or } x = 0.018$$

$\therefore$ No. of moles of NO_2 at equilibrium = $2x = 2 \times 0.018 = 0.036$ mole

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Equilibrium : Definition

1. In chemical reaction $A \rightleftharpoons B$, the system will be known in equilibrium when
- (a) A completely changes to B
(b) 50% of A changes to B
(c) The rate of change of A to B and B to A on both the sides are same
(d) Only 10% of A changes to B
2. According to law of mass action rate of a chemical reaction is proportional to
- (a) Concentration of reactants
(b) Molar concentration of reactants
(c) Concentration of products
(d) Molar concentration of products
3. The active mass of 64 gm of HI in a two litre flask would be
- (a) 2 (b) 1
(c) 5 (d) 0.25
4. The rate constant for forward and backward reactions of hydrolysis of ester are 1.1×10^{-2} and 1.5×10^{-3} per minute respectively. Equilibrium constant for the reaction is
- $$\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$$
- (a) 4.33 (b) 5.33
(c) 6.33 (d) 7.33

Equilibrium Constant : Expression

5. For the system $3A + 2B \rightleftharpoons C$, the expression for equilibrium constant is
- (a) $\frac{[3A][2B]}{C}$ (b) $\frac{[C]}{[3A][2B]}$
(c) $\frac{[A]^3[B]^2}{[C]}$ (d) $\frac{[C]}{[A]^3[B]^2}$
6. 4 moles of A are mixed with 4 moles of B. At equilibrium for the reaction $A + B \rightleftharpoons C + D$, 2 moles of C and D are formed. The equilibrium constant for the reaction will be

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 4

7. On a given condition, the equilibrium concentration of HI, H_2 and I_2 are 0.80, 0.10 and 0.10 mole/litre. The equilibrium constant for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ will be
- (a) 64 (b) 12
(c) 8 (d) 0.8
8. For the system $A(g) + 2B(g) \rightleftharpoons C(g)$, the equilibrium concentrations are (A) 0.06 mole/litre (B) 0.12 mole/litre (C) 0.216 mole/litre. The K_{eq} for the reaction is
- (a) 250 (b) 416
(c) 4×10^{-3} (d) 125
9. For the reaction $A + 2B \rightleftharpoons C$, the expression for equilibrium constant is

- (a) $\frac{[A][B]^2}{[C]}$ (b) $\frac{[A][B]}{[C]}$
(c) $\frac{[C]}{[A][B]^2}$ (d) $\frac{[C]}{2[B][A]}$

10. For which of the following reactions does the equilibrium constant depend on the units of concentration

- (a) $\text{NO}(g) \rightleftharpoons \frac{1}{2}\text{N}_2(g) + \frac{1}{2}\text{O}_2(g)$
(b) $\text{Zn}(s) + \text{Cu}^{2+}(aq) \rightleftharpoons \text{Cu}(s) + \text{Zn}^{2+}(aq)$
(c) $\text{C}_2\text{H}_5\text{OH}(l) + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(l) + \text{H}_2\text{O}(l)$
(Reaction carried in an inert solvent)
(d) $\text{COCl}_2(g) \rightleftharpoons \text{CO}(g) + \text{Cl}_2(g)$

11. Unit of equilibrium constant for the reversible reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ is
- (a) $\text{mol}^{-1} \text{ litre}$ (b) $\text{mol}^{-2} \text{ litre}$
(c) mol litre^{-1} (d) None of these

12. In a reaction $A + B \rightleftharpoons C + D$, the concentrations of A, B, C and D (in moles/litre) are 0.5, 0.8, 0.4 and 1.0 respectively. The equilibrium constant is
- (a) 0.1 (b) 1.0
(c) 10 (d) ∞
13. The suitable expression for the equilibrium constant of the reaction $2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$ is
- (a) $K_c = \frac{[2NOCl]}{[2NO][Cl_2]}$ (b) $K_c = \frac{[NOCl]^2}{[NO]^2[Cl_2]}$
(c) $K_c = \frac{[NOCl]^2}{[NO][Cl_2]^2}$ (d) $K_c = \frac{[NOCl]^2}{[NO]^2[Cl_2]^2}$
14. $A + B \rightleftharpoons C + D$. If finally the concentration of A and B are both equal but at equilibrium concentration of D will be twice of that of A then what will be the equilibrium constant of reaction.
- (a) 4/9 (b) 9/4
(c) 1/9 (d) 4
15. For the reaction $2SO_2 + O_2 \rightleftharpoons 2SO_3$, the units of K_c are
- (a) litre/mole (b) mol/litre
(c) $(\text{mol litre}^{-1})^2$ (d) $(\text{litre mole}^{-1})^2$
16. The equilibrium concentration of X, Y and YX_2 are 4, 2 and 2 moles respectively for the equilibrium $2X + Y \rightleftharpoons YX_2$. The value of K_c is
- (a) 0.625 (b) 0.0625
(c) 6.25 (d) 0.00625
17. In a 500 ml capacity vessel CO and Cl_2 are mixed to form $COCl_2$. At equilibrium, it contains 0.2 moles of $COCl_2$ and 0.1 mole of each of CO and CO_2 . The equilibrium constant K_c for the reaction $CO + Cl_2 \rightleftharpoons COCl_2$ is
- (a) 5 (b) 10
(c) 15 (d) 20
18. 4.5 moles each of hydrogen and iodine heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is
- (a) 1 (b) 10
(c) 5 (d) 0.33
19. An equilibrium mixture of the reaction $2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$ had 0.5 mole H_2S , 0.10 mole H_2 and 0.4 mole S_2 in one litre vessel. The value of equilibrium constant (K) in mole litre⁻¹ is
- (a) 0.004 (b) 0.008
(c) 0.016 (d) 0.160
20. For the equilibrium $N_2 + 3H_2 \rightleftharpoons 2NH_3$, K_c at 1000K is 2.37×10^{-3} . If at equilibrium $[N_2] = 2M$, $[H_2] = 3M$, the concentration of NH_3 is
- (a) 0.00358 M (b) 0.0358 M
(c) 0.358 M (d) 3.58 M
21. Two moles of NH_3 when put into a previously evacuated vessel (one litre), partially dissociate into N_2 and H_2 . If at equilibrium one mole of NH_3 is present, the equilibrium constant is
- (a) $3/4 \text{ mol}^2 \text{ litre}^{-2}$ (b) $27/64 \text{ mol}^2 \text{ litre}^{-2}$
(c) $27/32 \text{ mol}^2 \text{ litre}^{-2}$ (d) $27/16 \text{ mol}^2 \text{ litre}^{-2}$
22. In Haber process 30 litres of dihydrogen and 30 litres of dinitrogen were taken for reaction which yielded only 50% of the expected product. What will be the composition of gaseous mixture under the aforesaid condition in the end
- (a) 20 litres ammonia, 25 litres nitrogen, 15 litres hydrogen
(b) 20 litres ammonia, 20 litres nitrogen, 20 litres hydrogen
(c) 10 litres ammonia, 25 litres nitrogen, 15 litres hydrogen
(d) 20 litres ammonia, 10 litres nitrogen, 30 litres hydrogen
23. For the reaction equilibrium $N_2O_4 \rightleftharpoons 2NO_{2(g)}$, the concentrations of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol litre}^{-1}$ respectively. The value of K_c for the reaction is
- (a) $3.3 \times 10^2 \text{ mole litre}^{-1}$ (b) $3 \times 10^1 \text{ mol litre}^{-1}$
(c) $3 \times 10^{-3} \text{ mol litre}^{-1}$ (d) $3 \times 10^3 \text{ mol litre}^{-1}$
24. The equilibrium constant for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at temperature T is 4×10^{-4} . The value of K_c for the reaction $NO(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ at the same temperature is
- (a) 4×10^{-4} (b) 50
(c) 2.5×10^2 (d) 0.02

25. In the reaction, $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$. In a 2 litre flask 0.4 moles of each H_2 and I_2 are taken. At equilibrium 0.5 moles of HI are formed. What will be the value of equilibrium constant, K_c

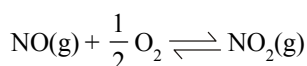
(a) 20.2 (b) 25.4
(c) 0.284 (d) 11.1

26. 2 mol of N_2 is mixed with 6 mol of H_2 in a closed vessel of one litre capacity. If 50% of N_2 is converted into NH_3 at equilibrium, the value of K_c for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ is

(a) 4/27 (b) 27/4
(c) 1/27 (d) 24

Equilibrium Constant : Effect of Stoichiometry

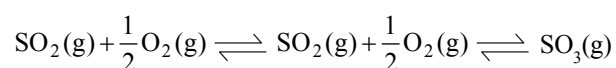
27. Equilibrium constants K_1 and K_2 for the following equilibria



and $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ are related as

(a) $K_2 = \frac{1}{K_1}$ (b) $K_2 = K_1^2$
(c) $K_2 = \frac{K_1}{2}$ (d) $K_2 = \frac{1}{K_1^2}$

28. Two gaseous equilibria



and

$2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ have equilibrium constants K_1 and K_2 respectively at 298 K. Which of the following relationships between K_1 and K_2 is correct

(a) $K_1 = K_2$ (b) $K_2 = K_1^2$
(c) $K_2 = 1/K_1^2$ (d) $K_2 = 1/K_1$

29. At a given temperature, the equilibrium constant for reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 2.4×10^{-3} . At the same temperature, the equilibrium constant for reaction $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ is

(a) 2.4×10^{-3} (b) -2.4×10^{-3}
(c) 4.2×10^2 (d) 4.8×10^{-2}

Position of Equilibrium : Extent of a Reaction

30. In which of the following, the reaction proceeds towards completion

(a) $K = 10^3$ (b) $K = 10^{-2}$
(c) $K = 10$ (d) $K = 1$

31. A 1 M solution of glucose reaches dissociation equilibrium according to equation given below $6\text{HCHO} \rightleftharpoons \text{C}_6\text{H}_{12}\text{O}_6$. What is the concentration of HCHO at equilibrium if equilibrium constant is 6×10^{22}

(a) $1.6 \times 10^{-8} \text{ M}$ (b) $3.2 \times 10^{-6} \text{ M}$
(c) $3.2 \times 10^{-4} \text{ M}$ (d) $1.6 \times 10^{-4} \text{ M}$

Kp and Kc Relationship

32. In which of the following reaction, the value of K_p will be equal to K_c

(a) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ (b) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
(c) $2\text{NH}_3 \rightleftharpoons \text{N}_2 + 3\text{H}_2$ (d) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$

33. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 + \text{heat}$

(a) $K_p = K_c (\text{RT})$ (b) $K_c = K_p (\text{RT})$
(c) $K_p = K_c (\text{RT})^{-2}$ (d) $K_p = K_c (\text{RT})^{-1}$

34. In which of the following equilibria, the value of K_p is less than K_c

(a) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$
(b) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
(c) $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$
(d) $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$

35. For the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ at 721K the value of equilibrium constant (K_c) is 50. When the equilibrium concentration of both is 0.5M, the value of K_p under the same conditions will be

(a) 0.002 (b) 0.2
(c) 50.0 (d) 50 / RT

36. The reaction between N_2 and H_2 to form ammonia has $K_c = 6 \times 10^{-2}$ at the temperature 500°C . The numerical value of K_p for this reaction is

(a) 1.5×10^{-5} (b) 1.5×10^5
(c) 1.5×10^{-6} (d) 1.5×10^6

37. For the reversible reaction,
- $$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
- at 500°C, the value of K_p is 1.44×10^{-5} when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mole litre⁻¹, is
- $1.44 \times 10^{-5}/(0.082 \times 500)^{-2}$
 - $1.44 \times 10^{-5}/(8.314 \times 773)^{-2}$
 - $1.44 \times 10^{-5}/(0.082 \times 773)^2$
 - $1.44 \times 10^{-5}/(0.082 \times 773)^{-2}$
- Equilibrium : Analysis**
38. A mixture of 0.3 mole of H_2 and 0.3 mole of I_2 is allowed to react in a 10 litre evacuated flask at 500°C. The reaction is $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, the K is found to be 64. The amount of unreacted I_2 at equilibrium is
- 0.15 mole
 - 0.06 mole
 - 0.03 mole
 - 0.2 mole
39. 0.6 mole of NH_3 in a reaction vessel of 2dm³ capacity was brought to equilibrium. The vessel was then found to contain 0.15 mole of H_2 formed by the reaction
- $$2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$$
- Which of the following statements is true
- 0.15 mole of the original NH_3 had dissociated at equilibrium
 - 0.55 mole of ammonia is left in the vessel
 - At equilibrium the vessel contained 0.45 mole of N_2
 - The concentration of NH_3 at equilibrium is 0.25 mole per dm³
40. 5 moles of SO_2 and 5 moles of O_2 are allowed to react to form SO_3 in a closed vessel. At the equilibrium stage 60% of SO_2 is used up. The total number of moles of SO_2 , O_2 and SO_3 in the vessel now is
- 10.0
 - 8.5
 - 10.5
 - 3.9
41. The equilibrium $\text{P}_4(\text{s}) + 6\text{Cl}_2(\text{g}) \rightleftharpoons 4\text{PCl}_3(\text{g})$ is attained by mixing equal moles of P_4 and Cl_2 in an evacuated vessel. Then at equilibrium
- $[\text{Cl}_2] > [\text{PCl}_3]$
 - $[\text{Cl}_2] > [\text{P}_4]$
 - $[\text{P}_4] > [\text{Cl}_2]$
 - $[\text{PCl}_3] > [\text{P}_4]$
42. 40% of a mixture of 0.2 mol of N_2 and 0.6 mol of H_2 reacts to give NH_3 according to the equation :
- $$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
- at constant temperature and pressure. Then the ratio of the final volume to the initial volume of gases are
- 4 : 5
 - 5 : 4
 - 7 : 10
 - 8 : 5
43. XY_2 dissociates as :
- $$\text{XY}_2(\text{g}) \rightleftharpoons \text{XY}(\text{g}) + \text{Y}(\text{g})$$
- Initial pressure of XY_2 is 600 mm Hg. The total pressure at equilibrium is 800 mm Hg. Assuming volume of system to remain constant, the value of K_p is
- 50
 - 100
 - 200
 - 400
44. 15 mol of H_2 and 5.2 moles of I_2 are mixed and allowed to attain equilibrium at 500°C. At equilibrium, the concentration of HI is found to be 10 mol. The equilibrium constant for the formation of HI is.
- 50
 - 25
 - 200
 - 15
45. For the equilibrium $\text{AB}(\text{g}) \rightleftharpoons \text{A}(\text{g}) + \text{B}(\text{g})$, K_p is equal to four times the total pressure. Calculate the number moles of A formed if one mol of AB is taken initially.
- 0.45
 - 0.30
 - 0.60
 - 0.90
46. $\text{I}_2 + \text{I}^- \rightleftharpoons \text{I}_3^-$
- This reaction is set-up in aqueous medium. We start with 1 mol of I_2 and 0.5 mol of I^- in 1L flask. After equilibrium is reached, excess of AgNO_3 gave 0.25 mol of yellow precipitate. Equilibrium constant is
- 1.33
 - 2.66
 - 2.00
 - 3.00
47. One mole of $\text{N}_2\text{O}_4(\text{g})$ at 100 K is kept in a closed container at 1.0 atm pressure. It is heated to 400 K, where 30% by mass of $\text{N}_2\text{O}_4(\text{g})$ decomposes of $\text{NO}_2(\text{g})$. The resultant pressure will be
- 4.2
 - 5.2
 - 3.2
 - 6.2

Equilibrium Constant and ΔG

48. $\Delta G^\circ (\text{HI}; \text{g}) \cong +1.7 \text{ kJ}$. What is the equilibrium constant at 25°C for $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
- (a) 24.0 (b) 3.9
(c) 2.0 (d) 0.5
49. Calculate ΔG° for conversion of oxygen to ozone $3/2 \text{ O}_2(\text{g}) \rightarrow \text{O}_3(\text{g})$ at 298 K , if K_p for this conversion is 2.47×10^{-29}
- (a) 163 kJ mol^{-1} (b) $2.4 \times 10^2 \text{ kJ mol}^{-1}$
(c) 1.63 kJ mol^{-1} (d) $2.38 \times 10^6 \text{ kJ mol}^{-1}$

Homogeneous and Heterogeneous Equilibria, Degree of Dissociation

50. Pure ammonia is placed in a vessel at temperature where its dissociation (α) is appreciable. At equilibrium
- (a) K_p does not change significantly with pressure
(b) α does not change with pressure
(c) Concentration of NH_3 does not change with pressure
(d) Concentration of H_2 is less than that of N_2
51. At a certain temp. $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$ Only 50% HI is dissociated at equilibrium. The equilibrium constant is
- (a) 0.25 (b) 1.0
(c) 3.0 (d) 0.50
52. 3.2 moles of hydrogen iodide were heated in a sealed bulb at 444°C till the equilibrium state was reached. Its degree of dissociation at this temperature was found to be 22%. The number of moles of hydrogen iodide present at equilibrium are
- (a) 2.496 (b) 1.87
(c) 2 (d) 4
53. $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ in closed container at equilibrium. What would be the effect of addition of CaCO_3 on the equilibrium concentration of CO_2 .
- (a) Increases (b) Decreases
(c) Data is not sufficient (d) Remains unaffected
54. An amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm. pressure. Ammonium hydrogen sulphide decomposes to yield NH_3 and H_2S gases in the flask. When the decomposition reaction reaches equilibrium, the total

pressure in the flask rises to 0.84 atm. The equilibrium constant for NH_4HS decomposition at this temperature is

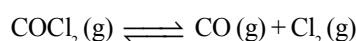
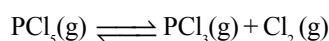
- (a) 0.30 (b) 0.18
(c) 0.17 (d) 0.11
55. Calculate the partial pressure of carbon monoxide from the following
- $\text{CaCO}_3(\text{s}) \xrightarrow{\Delta} \text{CaO}(\text{s}) + \text{CO}_2 \uparrow; K_p = 8 \times 10^{-2}$
 $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g}); K_p = 2$
- (a) 0.2 (b) 0.4
(c) 1.6 (d) 4
56. Change in volume of the system does not alter the number of moles in which of the following equilibrium
- (a) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
(b) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
(c) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
(d) $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$
57. The equilibrium constant (K_p) for the reaction $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 16. If the volume of the container is reduced to one half its original volume, the value of K_p for the reaction at the same temperature will be
- (a) 32 (b) 64
(c) 16 (d) 4
58. Consider the following equilibrium in a closed container : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
- At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements, holds true regarding the equilibrium constant (K_p) and degree of dissociation (α) ?
- (a) neither K_p nor α changes
(b) both V_p and α change
(c) K_p changes but α does not change
(d) K_p does not change but α changes
59. Calculate the partial pressure of carbon monoxide from the following data :
- $\text{CaCO}_3(\text{s}) \xrightarrow{\Delta} \text{CaO}(\text{s}) + \text{CO}_2 \uparrow, K_p = 8 \times 10^{-2}$
 $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \longrightarrow 2\text{CO}(\text{g}), K_p = 2$
- (a) 0.2 (b) 0.4
(c) 1.6 (d) 4
60. A small amount of NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. Ammonium hydrogen sulphide

decomposes to yield NH_3 and H_2S gases in the flask. When the decomposition reaction reaches equilibrium, the total pressure in the flask rises to 0.84 atm. The equilibrium constant for NH_4HS decomposition at this temperature is

- (a) 0.30 (b) 0.18
(c) 0.17 (d) 0.11

61. The vapour density of completely dissociated NH_4Cl is
(a) double than that of NH_4Cl
(b) half than that of NH_4Cl
(c) same as that of NH_4Cl
(d) determined by the amount of solid NH_4Cl taken

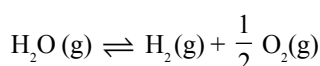
62. The following two equilibria exist simultaneously in a closed vessel :



If some CO is added into the vessel, then after the equilibrium is attained again, concentration of

- (a) PCl_5 will increase (b) PCl_5 will decrease
(c) PCl_5 will remain unaffected
(d) Cl_2 will increase

63. The equilibrium constant (K_p) for the decomposition of gaseous H_2O



is related to degree of dissociation (α) at a total pressure p , given by

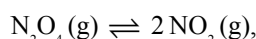
$$(a) K_p = \frac{\alpha^3 p^{1/2}}{(1+\alpha)(2+\alpha)^{1/2}}$$

$$(b) K_p = \frac{\alpha^3 p^{3/2}}{(1-\alpha)(2+\alpha)^{1/2}}$$

$$(c) K_p = \frac{\alpha^{3/2} p^2}{(1-\alpha)(2+\alpha)^{1/2}}$$

$$(d) K_p = \frac{\alpha^{3/2} p^{1/2}}{(1-\alpha)(2+\alpha)^{1/2}}$$

64. At the equilibrium of the reaction,

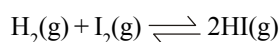


the observed molecular weight of N_2O_4 is 80 g mol^{-1} at 350 K. The percentage of dissociation of $\text{N}_2\text{O}_4(\text{g})$ at 350 K is

- (a) 10% (b) 15%
(c) 20% (d) 18%

Le Chatelier's Principle

65. What is the effect of halving the pressure by doubling the volume on the following system at 500°C



- (a) Shift to product side
(b) Shift to product formation
(c) Liquifaction of HI
(d) No effect

66. In equilibrium $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$

The equilibrium constant may change when

- (a) CH_3COO^- are added (b) CH_3COOH is added
(c) Catalyst is added (d) Mixture is heated

67. Which of the following statements regarding a chemical equilibrium is wrong

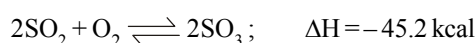
- (a) An equilibrium can be shifted by altering the temperature or pressure
(b) An equilibrium is dynamic
(c) The same state of equilibrium is reached whether one starts with the reactants or the products
(d) The forward reaction is favoured by the addition of a catalyst

68. In the reaction, $\text{A}_2(\text{g}) + 4\text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}_4(\text{g})$

$\Delta H < 0$ the formation of AB_4 is will be favoured at

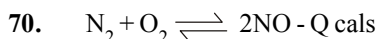
- (a) Low temperature, high pressure
(b) High temperature, low pressure
(c) Low temperature, low pressure
(d) High temperature, high pressure

69. The formation of SO_3 takes place according to the following reaction,



The formation of SO_3 is favoured by

- (a) Increasing in temperature
- (b) Removal of oxygen
- (c) Increase of volume
- (d) Increasing of pressure



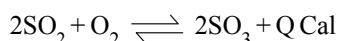
In the above reaction which is the essential condition for the higher production of NO

- (a) High temperature (b) High pressure
- (c) Low temperature (d) Low pressure

71. Which of the following reactions proceed at low pressure

- (a) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (b) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$
- (c) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ (d) $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$

72. In the following reversible reaction



Most suitable condition for the higher production of SO_3 is

- (a) High temperature and high pressure
- (b) High temperature and low pressure
- (c) Low temperature and high pressure
- (d) Low temperature and low pressure

73. The reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D} + \text{heat}$ has reached equilibrium. The reaction may be made to proceed forward by

- (a) Adding more C
- (b) Adding more D
- (c) Decreasing the temperature
- (d) Increasing the temperature

74. According to Le-chatelier principle, if heat is given to solid-liquid system, then

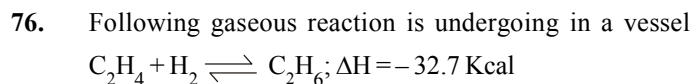
- (a) Quantity of solid will reduce
- (b) Quantity of liquid will reduce
- (c) Increase in temperature
- (d) Decrease in temperature

75. In the reaction $\text{A(g)} + 2\text{B(g)} \rightleftharpoons \text{C(g)} + Q \text{ kJ}$, greater product will be obtained or the forward reaction is favoured by

- (a) At high temperature and high pressure
- (b) At high temperature and low pressure

(c) At low temperature and high pressure

(d) At low temperature and low pressure



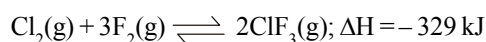
Which will increase the equilibrium concentration of C_2H_6

- (a) Increase of temperature
- (b) By reducing temperature
- (c) By removing some hydrogen
- (d) By adding some C_2H_6

77. The effect of increasing the pressure on the equilibrium $2\text{A} + 3\text{B} \rightleftharpoons 3\text{A} + 2\text{B}$ is

- (a) Forward reaction is favoured
- (b) Backward reaction is favoured
- (c) No effect
- (d) None of the above

78. The exothermic formation of ClF_3 is represented by the equation



Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2 , F_2 and ClF_3

- (a) Increasing the temperature
- (b) Removing Cl_2
- (c) Increasing the volume of the container
- (d) Adding F_2

79. In which of the following system, doubling the volume of the container cause a shift to the right

- (a) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{HCl}(\text{g})$
- (b) $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}_2(\text{g})$
- (c) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- (d) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

80. Which of the following information can be obtained on the basis of Le-chatelier's principle

- (a) Entropy change in a reaction
- (b) Dissociation constant of a weak acid
- (c) Equilibrium constant of a chemical reaction

- (d) Shift in equilibrium position on changing value of a constant
81. The equilibrium $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ is attained at 25°C in a closed container and an inert gas helium is introduced which of the following statement is correct
- (a) More chlorine is formed
 - (b) Concentration of SO_2 is reduced
 - (c) More SO_2Cl_2 is formed
 - (d) Concentration of $\text{SO}_2\text{Cl}_2(\text{SO}_2)$ and Cl_2 does not change
82. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ $\Delta H = +q$ cal, then formation of HI
- (a) Is favoured by lowering the temperature
 - (b) Is favoured by increasing the pressure
 - (c) Is unaffected by change in pressure
 - (d) Is unaffected by change in temperature
83. The formation of nitric oxide by contact process $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$. $\Delta H = 43.200$ kcal is favoured by
- (a) Low temperature and low pressure
 - (b) Low temperature and high pressure
 - (c) High temperature and high pressure
 - (d) High temperature and excess reactants concentration
84. The yield of product in the reaction $\text{A}_2(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{Q.kJ.}$ would be high at
- (a) High temperature and high pressure
 - (b) High temperature and low pressure
 - (c) Low temperature and high pressure
 - (d) Low temperature and low pressure

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

- In which of the following equilibrium change in the volume of the system does not alter the number of moles (2002)
 - $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
 - $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 - $\text{SO}_2\text{Cl}_2 \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$
- For the reaction (2002)

$$\text{CO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}), K_p/K_c \text{ is}$$
 - RT
 - $(RT)^{-1}$
 - $(RT)^{-1/2}$
 - $(RT)^{1/2}$
- In which of the following reactions, increase in the pressure at constant temperature does not affect the moles at equilibrium (2002)
 - $2\text{NH}_{3(\text{g})} \rightleftharpoons \text{N}_{2(\text{g})} + 3\text{H}_{2(\text{g})}$
 - $\text{C}_{(\text{g})} + 1/2\text{O}_{2(\text{g})} \rightleftharpoons \text{CO}_{(\text{g})}$
 - $\text{H}_{2(\text{g})} + 1/2\text{O}_{2(\text{g})} \rightleftharpoons \text{H}_2\text{O}_{(\text{g})}$
 - $\text{H}_{2(\text{g})} + \text{I}_{2(\text{g})} \rightleftharpoons 2\text{HI}_{(\text{g})}$
- For the reaction equilibrium (2003)

$$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$$
 the concentrations of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol L}^{-1}$ respectively. The value of K_c for the reaction is
 - $3 \times 10^3 \text{ mol L}^{-1}$
 - $3.3 \times 10^2 \text{ mol L}^{-1}$
 - $3 \times 10^{-1} \text{ mol L}^{-1}$
 - $3 \times 10^{-3} \text{ mol L}^{-1}$
- The conditions favourable for the reaction (2003)

$$2\text{SO}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{SO}_{3(\text{g})}; \quad \Delta H^\circ = -198 \text{ kJ}$$
 are
 - Low temperature, high pressure
 - Any value of T and P
 - Low temperature and low pressure
 - High temperature and high pressure
- What is the equilibrium expression for the reaction (2004)

$$\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightleftharpoons \text{P}_4\text{O}_{10}(\text{s}) ?$$
 - $K_c = \frac{[\text{P}_4\text{O}_{10}]}{[\text{P}_4][\text{O}_2]^5}$
 - $K_c = \frac{1}{[\text{O}_2]^5}$
 - $K_c = [\text{O}_2]^5$
 - $K_c = \frac{[\text{P}_4\text{O}_{10}]}{5[\text{P}_4][\text{O}_2]}$
- For the reaction (2004)

$$\text{CO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{COCl}_2(\text{g}),$$
 the $\frac{K_p}{K_c}$ is equal to
 - $\frac{1}{RT}$
 - 1.0
 - $\sqrt{RT}$
 - RT
- The equilibrium constant for the reaction (2004)

$$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$$
 at temperature T is 4×10^{-4} . The value of K_c for the reaction at the same temperature is
 - 2.5×10^2
 - 0.02
 - 4×10^{-4}
 - 50
- A small amount of NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. Ammonium hydrogen sulphide decomposes to yield NH_3 and H_2S gases in the flask. When the decomposition reaction reaches equilibrium, the total pressure in the flask rises to 0.84 atm. The equilibrium constant for NH_4HS decomposition at this temperature is (2005)
 - 0.30
 - 0.18
 - 0.17
 - 0.11

10. For the reaction
 $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$
 $(K_c = 1.8 \times 10^{-6} \text{ at } 184^\circ\text{C})$
 $(R = 0.0831 \text{ kJ/(mol.K)})$
 When K_p and K_c are compared at 184°C it is found that
 (2005)
 (a) Whether K_p is greater than, less than or equal of K_c depends upon the total gas pressure
 (b) $K_p = K_c$
 (c) K_p is greater than K_c
 (d) K_p is greater than K_c
11. Phosphorus pentachloride dissociates as follows, in a closed reaction vessel,
 $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ (2006)
 If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x , the partial pressure of PCl_3 will be
 (a) $\left(\frac{x}{x+1}\right)P$ (b) $\left(\frac{2x}{1-x}\right)P$
 (c) $\left(\frac{x}{x-1}\right)P$ (d) $\left(\frac{x}{1-x}\right)P$
12. The equilibrium constant for the reaction
 $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$
 is $K_c = 4.9 \times 10^{-2}$. The value of K_c for the reaction
 $2\text{SO}_2 + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ will be (2006)
 (a) 416 (b) 2.40×10^{-3}
 (c) 9.8×10^{-2} (d) 4.9×10^{-2}
13. The equilibrium constants K_{p1} and K_{p2} for the reactions
 $\text{X} \rightleftharpoons 2\text{Y}$ and $\text{Z} \rightleftharpoons \text{P} + \text{Q}$, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is (2008)
 (a) 1 : 36 (b) 1 : 1
 (c) 1 : 3 (d) 1 : 9
14. For the following three reactions 1, 2 and 3 equilibrium constants are given
 (1) $\text{CO}_{(\text{g})} + \text{H}_2\text{O}_{(\text{g})} \rightleftharpoons \text{CO}_{2(\text{g})} + \text{H}_{2(\text{g})}$; K_1
 (2) $\text{CH}_{4(\text{g})} + \text{H}_2\text{O}_{(\text{g})} \rightleftharpoons \text{CO}_{(\text{g})} + 3\text{H}_{2(\text{g})}$; K_2
 (3) $\text{CH}_{4(\text{g})} + 2\text{H}_2\text{O}_{(\text{g})} \rightleftharpoons \text{CO}_{2(\text{g})} + 4\text{H}_{2(\text{g})}$; K_3
 Which of the following relations is correct? (2008)
 (a) $K_1\sqrt{K_2} = K_3$ (b) $K_2K_3 = K_1$
 (c) $K_3 = K_1K_2$ (d) $K_3 = K_2^3K_1^2$
15. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K_p is (2011)
 (a) 1.8 atm (b) 3 atm
 (c) 0.3 atm (d) 0.18 atm
16. For the reaction $\text{SO}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightleftharpoons \text{SO}_{3(\text{g})}$, if $K_p = K_c(RT)^x$ where the symbols have usual meaning then the value of x is :
 (assuming ideality) (2014)
 (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
 (c) 1 (d) -1
17. The standard Gibbs energy change at 300 K for the reaction $2\text{A} \rightleftharpoons \text{B} + \text{C}$ is 2494.2 J. At a given time, the composition of the reaction mixture is $[\text{A}] = \frac{1}{2}$, $[\text{B}] = 2$ and $[\text{C}] = \frac{1}{2}$. The reaction proceeds in the : (2015)
 $[R = 8.314 \text{ J/K/mol}, e = 2.718]$
 (a) forward direction because $Q < K_c$
 (b) reverse direction because $Q < K_c$
 (c) forward direction because $Q > K_c$
 (d) reverse direction because $Q > K_c$
18. The equilibrium constant at 298 K for reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is 100. If the initial concentration of all the four species were 1M each, then equilibrium concentration of D (in mol L^{-1}) will be: (2016)
 (a) 0.818 (b) 1.818
 (c) 1.182 (d) 0.182
19. An aqueous solution contains 0.10 M H_2S and 0.20 M HCl . If the equilibrium constants for the formation of HS^- from H_2S is 1.0×10^{-7} and that of S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is: (2018)
 (a) 5×10^{-19} (b) 5×10^{-8}
 (c) 3×10^{-20} (d) 6×10^{-21}

JEE MAINS ONLINE QUESTION

1. At a certain temperature, only 50% HI is dissociated into H_2 and I_2 at equilibrium. The equilibrium constant is:

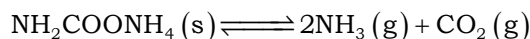
Online 2014 SET (1)

- (a) 1.0 (b) 3.0
(c) 0.5 (d) 0.25

2. What happens when an inert gas is added to an equilibrium keeping volume unchanged? Online 2014 SET (3)

- (a) More reactant will form
(b) More product will form
(c) Equilibrium will remain unchanged
(d) Less product will form

3. For the decomposition of the compound represented as



The $K_p = 2.9 \times 10^{-5} \text{ atm}^3$.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be :

Online 2014 SET (4)

- (a) $5.82 \times 10^{-2} \text{ atm}$ (b) $7.66 \times 10^{-2} \text{ atm}$
(c) $38.8 \times 10^{-2} \text{ atm}$ (d) $1.94 \times 10^{-2} \text{ atm}$

4. The increase of pressure on ice $\rightleftharpoons$ water system at constant temperature will lead to :

Online 2015 SET (1)

- (a) a decrease in the entropy of the system
(b) an increase in the Gibbs energy of the system
(c) no effects on the equilibrium
(d) a shift of the equilibrium in the forward direction

5. For the equilibrium, $A(g) \rightleftharpoons B(g)$, ΔH is -40 kJ/mol .

If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :

Online 2015 SET (1)

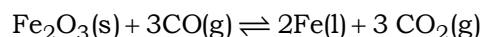
- (a) $E_f = 80 \text{ kJ/mol}$; $E_b = 120 \text{ kJ/mol}$
(b) $E_f = 60 \text{ kJ/mol}$; $E_b = 100 \text{ kJ/mol}$
(c) $E_f = 30 \text{ kJ/mol}$; $E_b = 70 \text{ kJ/mol}$
(d) $E_f = 70 \text{ kJ/mol}$; $E_b = 30 \text{ kJ/mol}$

6. A solid XY kept in an evacuated sealed container undergoes decomposition to form a mixture of gases X and Y at temperature T. The equilibrium pressure is 10 bar in this vessel. K_p for this reaction is :

Online 2016 SET (2)

- (a) 5 (b) 10
(c) 25 (d) 100

7. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal :



Using the Le Chatelier's principle, predict which one of the following will not disturb the equilibrium ?

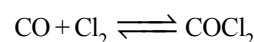
Online 2017 SET (2)

- (a) Removal of CO (b) Removal of CO_2
(c) Addition of CO_2 (d) Addition of Fe_2O_3

8. In which of the following reactions, an increase in the volume of the container will favour the formation of products ? Online 2018 SET (1)

- (a) $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$
(b) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(c) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(l)$
(d) $3O_2(g) \rightleftharpoons 2O_3(g)$

9. At a certain temperature in a 5L vessel, 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction,



At equilibrium, if one mole of CO is present then equilibrium constant (K_c) for the reaction is :

Online 2018 SET (2)

- (a) 2 (b) 2.5
(c) 3 (d) 4

10. The gas phase reaction $2NO_2(g) \rightarrow N_2O_4(g)$ is an exothermic reaction. The decomposition of N_2O_4 in equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$, can be increased by : Online 2018 SET (3)

- (a) lowering the temperature.
(b) increasing the pressure.
(c) addition of an inert gas at constant volume.
(d) addition of an inert gas at constant pressure.

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked "S" are single choice questions
2. All questions marked "M" are multiple choice questions
3. All questions marked "C" are comprehension based questions
4. All questions marked "A" are assertion–reason type questions
(A) If both assertion and reason are correct and reason is the correct explanation of assertion.
(B) If both assertion and reason are true but reason is not the correct explanation of assertion.
(C) If assertion is true but reason is false.
(D) If reason is true but assertion is false.
5. All questions marked "X" are matrix–match type questions
6. All questions marked "I" are integer type questions

Equilibrium : Definition and Expression

1. (A) **Assertion :** For a reaction at equilibrium, the free energy for the reaction is minimum.

Reason : The free energy for both reactants and products decreases and become equal.

- (a) A (b) B
(c) C (d) D

2. (A) **Assertion :** Chemical equilibrium represents a state of a reversible reaction in which measurable properties of the system (pressure, concentration, colour etc.) become constant under the given set of conditions.

Reason : The chemical equilibrium is an apparent state of rest in which two opposing reactions are proceeding at the same rate.

- (a) A (b) B
(c) C (d) D

3. (S) Which of the following statements about the reaction quotient, Q are correct ?

- (a) the reaction quotient, Q and the equilibrium constant always have the same numerical value
(b) Q may be lesser than, equal to or greater than K_{eq}
(c) Q (numerical value) varies as reaction proceeds
(d) $Q = 1$ at equilibrium

4. (S) For a reversible reaction : $A + B \rightleftharpoons C$

$$\left(\frac{dx}{dt}\right) = 2.0 \times 10^3 \text{ L mol}^{-1} \text{ s}^{-1} [A] [B] - 1.0 \times 10^2 \text{ s}^{-1} [C]$$

where x is the amount of 'A' dissociated. The value of equilibrium constant (K_{eq}) is :

- (a) 10 (b) 0.05
(c) 20 (d) Can't be calculated

5. (A) **Assertion :** The reaction quotient, Q has the same form as the equilibrium constant K_{eq} and is evaluated using any given concentrations of the species involved in the reaction, and not necessarily equilibrium concentrations.

Reason : If the numerical value of Q is not the same as the value of equilibrium constant, a reaction will proceed.

- (a) A (b) B
(c) C (d) D

6. (A) **Assertion :** The active mass of pure solids and pure liquids is taken unity.

Reason : The active mass of pure solids and liquids depends on density and molecular mass. The density and molecular mass of pure liquids and solids are constant.

- (a) A (b) B
(c) C (d) D

7. (A) **Assertion :** If some $\text{PCl}_5(\text{g})$ containing labelled phosphorus p^{31} is added to a system with following equilibrium, after sometime the system was found to contain radioactive PCl_3 .



Reason : Chemical equilibrium is dynamic in nature.

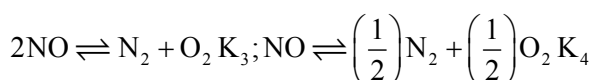
- (a) A (b) B
(c) C (d) D

8. (A) **Assertion :** As a reversible system approaches equilibrium, entropy of the system increases.

Reason : The state of equilibrium is the most disordered state of a reversible system.

- (a) A (b) B
(c) C (d) D

9. (M) $\text{N}_2\text{O}_2 \rightleftharpoons 2\text{KOK}_1; \left(\frac{1}{2}\right)\text{N}_2 + \left(\frac{1}{2}\right)\text{O}_2 \rightleftharpoons \text{NOK}_2$



Correct relation(s) between K_1 , K_2 , K_3 and K_4 is/are :

- (a) $\text{K}_2 \times \text{K}_4 = 1$ (b) $\sqrt{\text{K}_1 \times \text{K}_4} = 1$
(c) $\sqrt{\text{K}_3} \times \text{K}_2 = 1$ (d) $\sqrt{\text{K}_3 \times \text{K}_2} = 1$

K_p and K_c

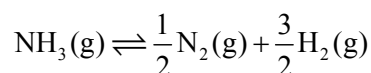
10. (X) Match the following systems with their K_p/K_c relationship

Equilibrium types

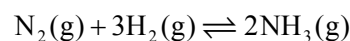
**K_p/K_c
relationship**

- (a) $\text{NO}(\text{g}) + \text{NO}_3(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ (p) $\text{K}_p > \text{K}_c$
(b) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ (q) $\text{K}_p = \text{K}_c(\text{RT})^2$
(c) $2\text{O}_3(\text{g}) \rightleftharpoons 3\text{O}_2(\text{g})$ (r) $\text{K}_p = \text{K}_c$
(d) $\text{BrF}_5(\text{g}) \rightleftharpoons \frac{1}{2}\text{Br}_2(\text{g}) + \frac{5}{2}\text{F}_2(\text{g})$ (s) $\text{K}_p < \text{K}_c$

11. (S) Equilibrium constant K_c for the following reaction at 800K is, 4.



The value of K_p for the following reaction will be :



- (a) $\left(\frac{800\text{R}}{4}\right)^{-2}$ (b) $16 \times (800\text{R})^2$
(c) $\left[\frac{1}{4 \times 800\text{R}}\right]^2$ (d) $(800\text{R})^{1/2} 4$

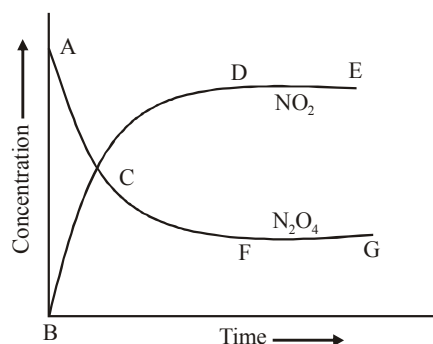
Equilibrium : Analysis

12. (S) $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$, $\text{K}_c = 4$. This reversible reaction is studied graphically as shown in the given figure. Select the correct statement out of I, II and III.

I : Reaction quotient has maximum value at point A

II : Reaction proceeds left to right at a point when $[\text{N}_2\text{O}_4] = [\text{NO}_2] = 0.1 \text{ M}$

III : $\text{K}_c = Q$ when point D or F is reached :



- (a) I, II (b) II, III
(c) II (d) I, II, III

13. (S) $\text{I}_2 + \text{I}^- \rightleftharpoons \text{I}_3^-$

This reaction is set-up in aqueous medium. We start with 1 mol of I_2 and 0.5 mol of I^- in 1L flask. After equilibrium is reached, excess of AgNO_3 gave 0.25 mol of yellow ppt. Equilibrium constant is : [Given : AgNO_3 gives yellow ppt with I^-]

- (a) 1.33 (b) 2.66
(c) 2.00 (d) 3.00

- 14. (I)** Given the hypothetical reaction :
 $2A(s) + nB(g) \rightleftharpoons 3C(g)$ $K_p = 0.0105$
 and $K_c = 0.45$ at 250°C . What is the value of coefficient 'n' ?
- 15. (S)** In the reaction : $A + B \rightleftharpoons 2C + D$. The initial concentration of A and B are 1M each. The value of K_c is 10^8 . What is the equilibrium concentration of A ?
 (a) 2×10^{-4} M (b) 2×10^4 M
 (c) 0.005 M (d) 0.0025 M
- 16. (S)** Equilibrium constant for two complexes are :
 A : $K_4[\text{Fe}(\text{CN})_6]$ 2.6×10^{37} (for dissociation)
 B : $K_3[\text{Fe}(\text{CN})_6]$ 1.9×10^{17} (for dissociation)
 (a) A and B are equally stable
 (b) A is more stable than A
 (c) B is more stable than A
 (d) the predictable stability
- 17. (S)** For the reaction, $[\text{Ag}(\text{CN})_2]^- \rightleftharpoons \text{Ag}^+ + 2\text{CN}^-$, the equilibrium constant K_c at 25°C is 4×10^{-19} ; then Ag^+ concentration in a solution which has 0.1 M KCN and 0.03M AgNO_3 is :
 (a) 7.5×10^{18} (b) 7.5×10^{-18}
 (c) 7.5×10^{19} (d) 7.5×10^{-19}
- 18. (S)** A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted to CO on addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :
 (a) 0.18 atm (b) 1.8 atm
 (c) 3 atm (d) 0.3 atm
- 19. (S)** The equilibrium reaction : $2AB(aq) \rightleftharpoons 2A(aq) + B_2(g)$ has an equilibrium constant equal to 1.28. If the concentrations of AB and A are 5 mol/L and 4 mol/L respectively, what is $[B_2]$?
 (a) 0.50 M (b) 1.60 M
 (c) 2.00 M (d) 2.84 M
- 20. (S)** Consider the reaction $\text{NaBr}(aq) + \text{H}_2\text{SO}_4(aq) \rightleftharpoons \text{NaHSO}_4(aq) + \text{HBr}(aq)$. The equilibrium constant is 8.3×10^{-2} . If the equilibrium concentrations of NaBr, H_2SO_4 and NaHSO_4 are 6.0 M, 9.0 M and 3.0 M respectively, what is the molarity of HBr ?
 (a) 1.8×10^{-2} M (b) 6.7×10^{-1} M
 (c) 1.5×10^0 M (d) 3.6×10^1 M
- 21. (S)** For the all gas reaction at 1000 K : $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$, $K = 0.0344$. If you were to begin with 8.0 M of HI in a fixed volume container, what would be the equilibrium concentration of H_2 ?
 (a) 1.10 M (b) 0.48 M
 (c) 5.8 M (d) 2.92 M
- 22. (S)** In reversible reaction $A \xrightleftharpoons[k_2]{k_1} B$, the initial concentration of A and B are a and b in moles per litre and the equilibrium concentration are (a - x) and (b - x) respectively ; express x in terms of k_1 , k_2 , a and b.
 (a) $\frac{k_1 a - k_2 b}{k_1 + k_2}$ (b) $\frac{k_1 a - k_2 b}{k_1 - k_2}$
 (c) $\frac{k_1 a - k_2 b}{k_1 k_2}$ (d) $\frac{k_1 a + k_2 b}{k_1 + k_2}$
- Equilibrium Constant and ΔG**
- 23. (S)** If for the heterogeneous equilibrium
 $\text{CaCO}_3(s) \rightleftharpoons \text{CaO}(s) + \text{CO}_2(g)$; $K = 1$ at 1 atm, the temperature is given by :
 (a) $T = \frac{\Delta S^\circ}{\Delta H^\circ}$ (b) $T = \frac{\Delta H^\circ}{\Delta S^\circ}$
 (c) $T = \frac{\Delta G^\circ}{R}$ (d) $T = \frac{\Delta G^\circ}{\Delta H^\circ}$
- 24. (X)** Match the Column-I with Column-II :

Column-I (Equations)	Column-II (Type of process)
(a) $K_p > Q$	(p) Non spontaneous
(b) $\Delta G^\circ < RT \log_e Q$	(q) Equilibrium
(c) $K_p = Q$	(r) Spontaneous and endothermic
(d) $T > \frac{\Delta H}{\Delta S}$	(s) Spontaneous

Passage (For Questions 25 to 27)

Effect of temperature on the equilibrium process is analyzed by using the thermodynamics. From the thermodynamics relation.

$$\Delta G^\circ = -2.303 RT \log K \quad \dots(i)$$

ΔG° : Standard free energy change

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \quad \dots(ii)$$

ΔH° : Standard heat of the reaction

From (i) and (ii)

$$-2.303 RT \log K = \Delta H^\circ - T\Delta S^\circ$$

ΔS° : Standard entropy change

$$\Rightarrow \log K = -\frac{\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R} \quad \dots(iii)$$

If a plot of $\log K$ vs $1/T$ is made then it is a straight line

$$\text{having slope} = \frac{-\Delta H^\circ}{2.303R} \text{ and Y intercept} = \frac{\Delta S^\circ}{2.303R}$$

If at temperature T_1 , equilibrium constant be K_1 and at temperature T_2 , equilibrium constant be K_2 then the above equation reduces to :

$$\Rightarrow \log K_1 = -\frac{\Delta H^\circ}{2.303RT_1} + \frac{\Delta S^\circ}{2.303R}$$

$$\Rightarrow \log K_2 = -\frac{\Delta H^\circ}{2.303RT_2} + \frac{\Delta S^\circ}{2.303R}$$

Subtracting (iv) from (v) we get.

$$\Rightarrow \log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

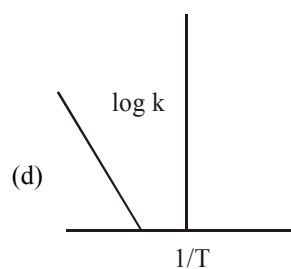
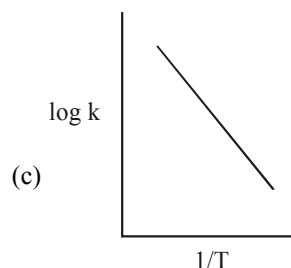
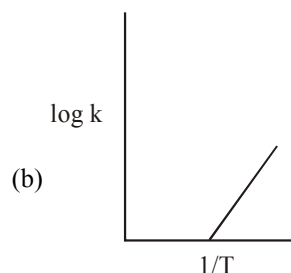
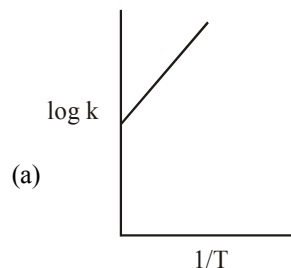
From the above relation we can conclude that the value of equilibrium constant increases with increase in temperature for an endothermic reaction and the same decreases with the increase in temperature for an exothermic reaction. Answer the following three questions based on the above information.

25. (C) If standard heat of dissociation of PCl_5 is 230 cal, then

slope of the graph of $\log k$ vs $\frac{1}{T}$ is

- (a) +50 (b) -50
(c) 10 (d) None of these

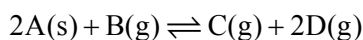
26. (C) For exothermic reaction is $\Delta S^\circ < 0$, then the sketch of $\log K$ vs $1/T$ may be



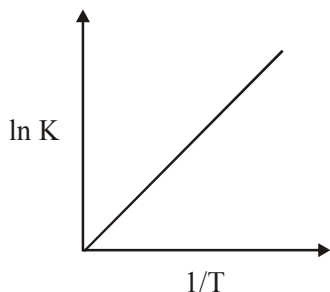
27. (C) If for a particular reversible reaction : $K_c = 57$ at 355°C and $K_c = 69$ at 450° , then

- (a) $\Delta H < 0$
(b) $\Delta H > 0$
(c) $\Delta H = 0$
(d) Sign of ΔH can't be determined

28. (M) Variation of equilibrium constant K for the reaction



is plotted against absolute temperature T in figure as :
ln K Vs (1/T)



ln K vs 1/T

- (a) the forward reaction is exothermic
- (b) the forward reaction is endothermic
- (c) the slope of line is proportional to ΔH
- (d) adding 'A' favours forward reaction

29. (M) Which is/are correct ?

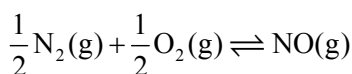
(a) $2.303 \log K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$

(b) $\Delta G^\circ = -2.303 RT \log K$

(c) $-2.303 \log K = -\frac{\Delta H^\circ}{RT^2} + \frac{\Delta S^\circ}{R}$

(d) $2.303 \log K = \frac{1}{RT}(\Delta H^\circ + \Delta S^\circ)$

30. (S) The free energy of formation of NO is 78 kJ mol^{-1} at the temperature of an automobile engine (1000 K). What is the equilibrium constant for this reaction at this reaction at 1000 K ?



- (a) 8.4×10^{-5}
- (b) 7.1×10^{-9}
- (c) 4.2×10^{-10}
- (d) 1.7×10^{-19}

31. (M) The variation of equilibrium constant K with temperature is represented by :

(a) $\log_e K_2 - \log_e K_1 = -\frac{\Delta H}{R} \int_{T_1}^{T_2} \left(\frac{1}{T} \right) dT$

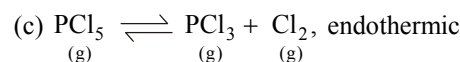
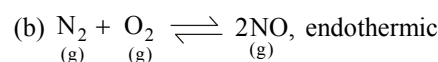
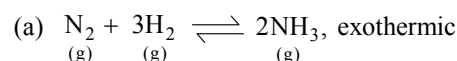
(b) $\left\{ \frac{d \log_e K}{dT} \right\}_p = -\frac{\Delta H}{RT}$

(c) $\log_e K = \text{Constant} - \frac{RT}{\Delta H}$

(d) $\log_e K = \text{Constant} - \frac{\Delta H}{RT}$

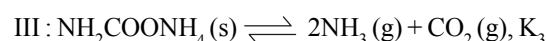
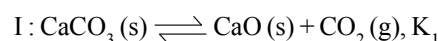
Homogeneous and Heterogeneous Equilibria, Degree of Dissociation

32. (S) Volume of the flask in which the following equilibria are separately established are transferred to a flask that is double the size of the earlier flask. In which of the following cases, equilibrium constants are affected ?



(d) none of these

33. (S) In each of the following, total pressure set-up at equilibrium is assumed to be equal and is one atm with equilibrium constants K_p given :



In the increasing order :

- (a) $K_1 = K_2 = K_3$
- (b) $K_1 < K_2 < K_3$
- (c) $K_3 < K_2 < K_1$
- (d) None of these

34. (S) During thermal dissociation, the observed vapour density of $N_2O_4(g)$ is 26.0. The extent of dissociation of $N_2O_4(g)$ is :

- (a) 50%
- (b) 87%
- (c) 77%
- (d) 23%

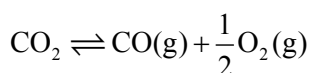
35. (S) For the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, the value of K_p is 1.7×10^3 at 500 K and 1.78×10^4 at 600 K. Which of the following is correct ?

- The proportion of NO_2 in the equilibrium mixture is increased by decreasing the pressure
- The standard enthalpy change for the forward reaction is negative
- units of K_p are atm
- at 500 K, the degree of dissociation N_2O_4 decreases by 50% by increasing the pressure by 100%.

The correct choice is :

- (a) I, III (b) I
(c) II, IV (d) III

36. (S) Consider the equilibrium



The equilibrium constant K is given by (when $\alpha \ll 1$) :

- (a) $K = \frac{\alpha^{3/2}}{\sqrt{2}}$ (b) $K = \frac{\alpha^3}{2}$
(c) $K = \frac{\alpha^{3/2}}{2}$ (d) $K = \frac{\alpha^{3/2}}{\sqrt{3}}$

37. (X) Match the Column-I with Column-II :

Column-I

(Reaction)

Column-II

(Effect on equilibrium)

- (A) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3$ (q) Forward shift by lowering of
 $\Delta H = -ve$ temperature
- (B) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ (q) Forward shift by rise in
 $\Delta H = +ve$ temperature
- (c) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ (r) No effect of pressure
 $\Delta H = +ve$ change
- (d) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ (s) Forward shift by rise in
 $\Delta H = -ve$ pressure

38. (I) For the equilibrium $\text{AB}(\text{g}) \rightleftharpoons \text{A}(\text{g}) + \text{B}(\text{g})$, at a given temperature $\frac{1}{3}$ rd of AB is dissociated, then $\frac{P}{K_p}$ will be numerically equal to _____.

39. (I) SO_2Cl_2 and Cl_2 are introduced into a 3L vessel. Partial pressure of SO_2Cl_2 and Cl_2 at equilibrium are 1 atm and 2 atm respectively. The value of K_p for the following reaction



The total pressure in atm at equilibrium would be _____.

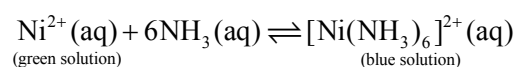
40. (I) What weight of solid ammonium carbamate ($\text{NH}_2\text{COONH}_4$) when vaporised at 200°C will have a volume of 8.96 litre at 1 atm ? Assume that solid completely decomposes into CO_2 and NH_3 at 200°C and 1 atm.

41. (A) **Assertion :** The dissociation of CaCO_3 can be represented as, $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. Some solid CaCO_3 is placed in an evacuated vessel enclosed by a piston and heated so that a portion of it is decomposed. If the piston is moved so that the volume of the vessel is doubled, while the temperature is held constant, the number of moles of CO_2 in the vessel increases.

Reason : The pressure of CO_2 in the vessel will remain the same.

Passage : For Questions (42 to 44)

These 3 questions deal with the following chemical reaction:



When $\text{H}^+(\text{aq})$ is added, the colour green is favoured. Use one or more of the following interpretations to answer the questions.

- Some unreacted $\text{Ni}^{2+}(\text{aq})$ is present in the solution at equilibrium.
- Some unreacted $\text{NH}_3(\text{aq})$ is present in the solution at equilibrium.

III. The colour change indicates new equilibrium conditions with reduced $[\text{Ni}(\text{NH}_3)_6]^{2+}(\text{aq})$.

IV. The colour change indicates new equilibrium conditions with increased $[\text{Ni}(\text{NH}_3)_6]^{2+}(\text{aq})$.

42. (C) The deepening colour on dissolving more $\text{Ni}(\text{NO}_3)_2$ supports interpretation(s)

- (a) I only (b) I and IV only
(c) II and IV only (d) I and II only

43. (C) The deepening colour on addition of more $\text{NH}_3(\text{aq})$ supports interpretation(s)

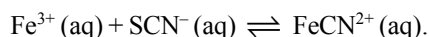
- (a) I only (b) I and IV only
(c) I and II only (d) II and IV only

44. (C) The disappearance of colour on addition of $\text{H}^+(\text{aq})$ supports interpretation(s)

- (a) I only (b) II and IV only
(c) I and II only (d) III only

Passage : For Questions (45 to 47)

These 3 questions refer to the following experiment. Some $\text{Fe}(\text{NO}_3)_3$ solution is added to KSCN solution and a dark red colour appears. This colour is caused by FeSCN^{2+} , which is formed by the reaction :



Experiments show this colour becomes deeper upon addition of more $\text{Fe}(\text{NO}_3)_3$ or KSCN. The colour disappears when Na_2HPO_4 is added. Use one or more of the interpretations below.

- (a) Unreacted $\text{Fe}^{3+}(\text{aq})$ is present in the original equilibrium mixture.
(b) Unreacted $\text{SCN}^-(\text{aq})$ is present in the original equilibrium mixture.
(c) Colour change indicates a new equilibrium ; $[\text{FeSCN}^{2+}(\text{aq})]$ has been reduced.
(d) Colour change indicates a new equilibrium ; $[\text{FeSCN}^{2+}(\text{aq})]$ has been increased.

45. (C) The deepening of colour on addition of more $\text{Fe}(\text{NO}_3)_3$ supports

- (a) A only (b) A and D only
(c) A and B only (d) B and D only

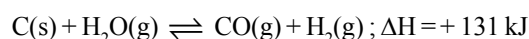
46. (C) The deepening of colour on addition of KSCN supports

- (a) A only (b) A and D only
(c) A and B only (d) B and D only

47. (C) The disappearance of colour on addition of Na_2HPO_4 supports

- (a) A only (b) A and D only
(c) A and B only (d) C only

48. (M) An industrial fuel, 'water gas' which consists of a mixture of H_2 and CO can be made by passing steam over red-hot carbon. The reaction is



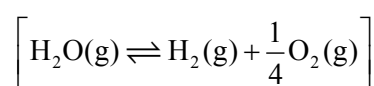
The yield of CO and H_2 at equilibrium would be shifted to the product side by

- (a) raising the relative pressure of steam
(b) adding hot carbon
(c) raising the temperature
(d) reducing the volume of the system

49. (S) For the reaction, $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$; if percentage dissociation of N_2O_4 are 25%, 50%, 75% and 100%, then the sequence of observed vapour densities will be :

- (a) $d_1 > d_2 > d_3 > d_4$ (b) $d_4 > d_3 > d_2 > d_1$
(c) $d_1 = d_2 = d_3 = d_4$ (d) $(d_1 = d_2) > (d_3 = d_4)$

50. (S) The equilibrium constant for the decomposition of water,



is given by :

$$(a) K = \frac{\alpha^3 p^{1/2}}{(1-\alpha)(2-\alpha)^{1/2}}$$

$$(b) K = \frac{\alpha^{3/2} p^{1/2}}{(1-\alpha)(2+\alpha)^{1/2}}$$

$$(c) K = \frac{\alpha^3 p^{1/2}}{\sqrt{2}}$$

$$(d) K = \frac{\alpha^3 p^{3/2}}{(1-\alpha)(2+\alpha)^{1/2}}$$

51. (S) The equilibrium constants K_{p_1} and K_{p_2} for the reactions $X \rightleftharpoons 2Y$ and $Z \rightleftharpoons P + Q$; respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal, then the ratio of total pressures at these equilibria is :

- (a) 1 : 9 (b) 1 : 36
(c) 1 : 1 (d) 1 : 3

52. (S) In the preparation of CaO from CaCO_3 using the equilibrium, $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, (At 1 atm)

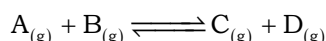
$$K_p \text{ is expressed as } \log K_p = 7.282 - \frac{8500}{T}.$$

For complete decomposition of CaCO_3 the temperature in celsius to be used is :

- (a) 1167 (b) 894
(c) 8500 (d) 850

Le Chatelier's Principle

53. (A) **Assertion :** For the reaction



at the given temperature, there will be no effect by addition of inert gas either at constant pressure or at constant volume.

Reason : For the reaction where $\Delta n = 0$, there is no effect of inert gas either at constant volume or at constant pressure because K_p becomes purely a number.

- (a) A (b) B
(c) C (d) D

54. (A) **Assertion :** Addition of an inert gas at constant pressure equilibrium will support the dissociation of PCl_5 at the same temperature.

Reason : The addition of an inert gas will tend to increase the total pressure.

- (a) A (b) B
(c) C (d) D

55. (A) **Assertion :** The addition of an inert gas under constant volume condition cause the total pressure of the system to increase. This would thus lower the degree of dissociation of PCl_5 .

Reason : Addition of inert gas at equilibrium increase the kinetic energy of molecules.

- (a) A (b) B
(c) C (d) D

56. (M) Yield of NH_3 in Haber's process



can be increased by

- (a) compressing the reaction system
(b) raising the temperature
(c) decreasing the temperature
(d) using the catalyst to lower down the activation energy

57. (M) The volume of the reaction flask is reduced to half of its initial value, temperature being constant. In which of the following cases the position of the equilibrium would shift ?

- (a) $\text{NH}_4\text{HS}_{(\text{s})} \rightleftharpoons \text{NH}_{3(\text{g})} + \text{H}_2\text{S}_{(\text{s})}$
(b) $2\text{NOCl}_{(\text{g})} \rightleftharpoons 2\text{NO}_{(\text{g})} + \text{Cl}_{2(\text{g})}$
(c) $\text{CO}_{(\text{g})} + \text{H}_2\text{O}_{(\text{g})} \rightleftharpoons \text{CO}_{2(\text{g})} + \text{H}_{2(\text{g})}$
(d) $\text{I}_{2(\text{g})} \rightleftharpoons 2\text{I}_{(\text{g})}$

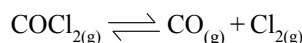
58. (M) $\text{KNO}_{3(\text{s})}$ dissociates on heating as :



At equilibrium in a closed container

- (a) addition of $\text{NaNO}_{3(\text{s})}$ favours forward reaction
(b) addition of $\text{KNO}_{2(\text{s})}$ favours reverse reaction
(c) increasing temperature favours forward reaction
(d) decreasing pressure favours forward reaction

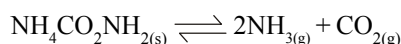
59. (M) The dissociation of phosgene, which occurs according to the reaction :



is an endothermic process. Which of the following factors will increase the degree of dissociation of COCl_2 ?

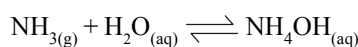
- (a) Adding Cl_2 to the system
(b) Adding helium to the system at constant pressure
(c) Decreasing the temperature of the system
(d) Reducing the total pressure

60. (M) The dissociation of ammonium carbamate may be represented by the equation

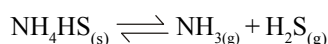


ΔH° for the forward reaction is negative. The equilibrium will shift from right to left if there is

- (a) a decrease in pressure
(b) an increase in temperature
(c) an increase in the concentration of ammonia
(d) an increase in the concentration of carbon dioxide
61. (M) Which of the following factors will increase solubility of a well known weak base, $\text{NH}_3(\text{g})$, in H_2O ?

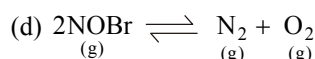
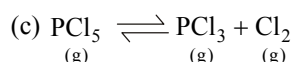
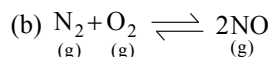
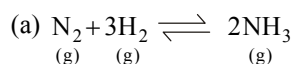


- (a) increase in pressure (b) addition of water
(c) addition of an acid to water
(d) addition of a base to water
62. (M) For the following endothermic equilibrium established by dissociation of NH_4HS to its components,



partial pressure of NH_3 will increase

- (a) if NH_3 is added after equilibrium is established
(b) if H_2S is added after equilibrium is established
(c) temperature is increased
(d) volume of the flask is decreased
63. (M) Volume of the flask in which the following equilibria are separately established are transferred to a flask that is double the size of the earlier flask. In which of the following cases, equilibrium concentrations are affected?

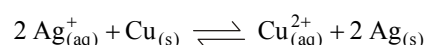


64. (S) To the system $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

in equilibrium, some N_2 gas was added at constant volume. Then,

- (a) K_p will remain constant and K_c will change
(b) K_c will remain constant and K_p will change
(c) Both K_p and K_c will remain constant
(d) Both K_p and K_c will change

65. (S) In a 1.0 L aqueous solution when the reaction



reaches equilibrium, $[\text{Cu}^{2+}] = x \text{ M}$ and $[\text{Ag}^+] = y \text{ M}$.

If the volume of solution is doubled by adding water, then at equilibrium:

(a) $[\text{Cu}^{2+}] = \frac{x}{2} \text{ M}$, $[\text{Ag}^+] = \frac{y}{2} \text{ M}$

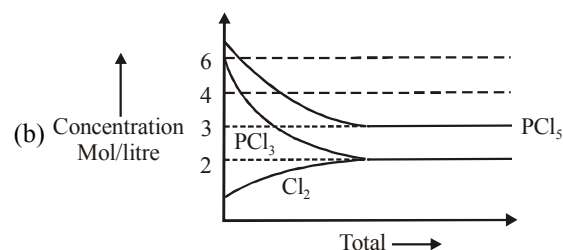
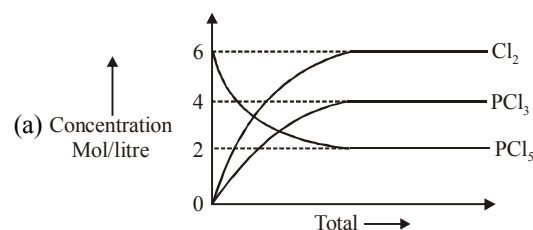
(b) $[\text{Cu}^{2+}] > \frac{x}{2} \text{ M}$, $[\text{Ag}^+] > \frac{y}{2} \text{ M}$

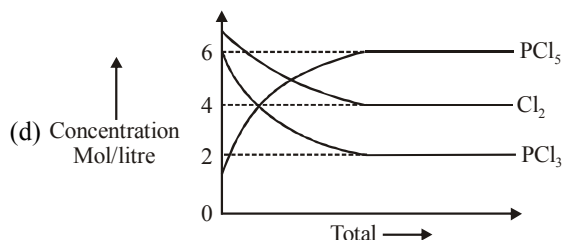
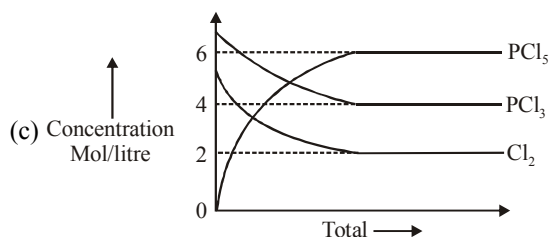
(c) $[\text{Cu}^{2+}] < \frac{x}{2} \text{ M}$, $[\text{Ag}^+] > \frac{y}{2} \text{ M}$

(d) $[\text{Cu}^{2+}] < \frac{x}{2} \text{ M}$, $[\text{Ag}^+] < \frac{y}{2} \text{ M}$

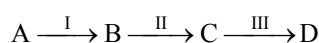
66. (S) For the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

Which of the following sketch may represent above equilibrium? Assume equilibrium can be achieved from either side and by taking any one or more components initially. (Given K_c for the reaction < 2)

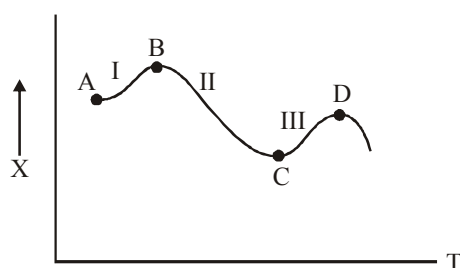




67. (S) For the following reaction through stages I, II and III



quantity of the product formed (x) varies with temperature (T) as given. Select correct statement :



- (a) Stages I and III are endothermic but II is exothermic
 (b) Stages I and III are exothermic but II is endothermic
 (c) Stages II and III are exothermic but I is endothermic
 (d) Stage I is exothermic but stages II and III are endothermic

68. (X) Match the conditions on the left column with states on the right column.

Equilibrium types

K_p/K_c
relationship

(a) $A_2(g) \rightleftharpoons 2A(g)$; $K_c = 1$.
Mixture of 1.0 mole of each
is prepared in a 1.0 L flask.

(p) System is at
equilibrium

(b) $A(g) + 2B(g) \rightleftharpoons AB_2(g)$
 $K_c = 2$. A mixture of 2.0 moles
of each A, B and AB_2
is prepared in a 1.0 L flask.

(q) Goes to right

(c) $A_2(g) \rightleftharpoons 2A(g)$; $K_p = 2$. At
equilibrium the total pressure
is 2.0 atm.

(r) $n(A) > n(A_2)$ at
equilibrium

(d) $2A(g) \rightleftharpoons A_2(g)$; $K_p = 2$. A
mixture, initially containing
both A and A_2 at 1.0 atm
partial pressure is prepared.

(s) $n(A) < n(A_2)$ at
equilibrium

69. (X) Match the Column-I with Column-II

Column-I

Column-II

(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(p) Forward shift by
increase in pressure

(B) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(p) No effect of
pressure change

(C) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(r) No effect of inert
gas addition at
constant volume

(D) $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g)$
+ $CO_2(g)$

(s) Addition of inert gas
will favour backward
direction (Not at
constant volume)

Passage : For Questions (70 to 72)

Le-Chatelier's principle : If a system at equilibrium is subjected to a change of any one of the factors such as concentration, pressure or temperature, the system adjusts itself in such a way as to "Nullify" the effect of that change.

Change of Pressure :

If a system in equilibrium consists of gases, then the concentrations of all components can be altered by changing the pressure. When the pressure on the system is increased, the volume decreases proportionately. The total number of moles per unit volume will now be more and the equilibrium will shift in that direction in which there is decrease in number of moles i.e., towards the direction in which there is decrease in volume.

Effect of Pressure on Melting Point :

There are two types of solids :

(a) Solids whose volume decreases on melting e.g., ice, diamond, carborundum, magnesium nitride and quartz.

Solid (higher volume) $\rightleftharpoons$ liquid (lower volume)

The process of melting is facilitated at higher pressure, thus the melting point is lowered.

(b) Solids whose volume increase on melting e.g. Fe, Cu, Ag, Au, etc.

Solid (lower volume) $\rightleftharpoons$ liquid (higher volume)

In this case, the process of melting becomes difficult at high pressure ; thus melting point becomes high.

(c) Solubility of substances : When solid substances are dissolved in water, either heat is evolved (exothermic) or heat is absorbed (endothermic) :



In such cases, solubility increases with increase in temperature. Consider the case of KOH; when this is dissolved, heat is evolved;



In such cases, solubility decrease with increase in temperature.

(d) Solubility of gases in liquids : When a gas dissolves in liquid, there is decrease in volume. Thus, increase of pressure will favour the dissolution of gas in liquid. Answer the following three questions based on the above information.

70. (C) When a gas "X" is dissolved in water, the heat is evolved. Then solubility of 'X' will increase at

- (a) low pressure, high temperature
- (b) low pressure, low temperature
- (c) high pressure, high temperature
- (d) high pressure, low temperature

71. (C) $\text{Au(s)} \rightleftharpoons \text{Au(l)}$

Above equilibrium is favoured at

- (a) high pressure low temperature
- (b) high pressure high temperature
- (c) low pressure, high temperature
- (d) low pressure, low temperature

72. (C) For the reaction : $\frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{NO}(\text{g})$

If pressure is increased by reducing the volume of the container, then

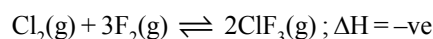
- (a) total pressure at equilibrium will change
- (b) concentration of all the components at equilibrium will change
- (c) concentration of all the components at equilibrium will remain same
- (d) equilibrium will shift in the forward direction

73. (M) For the reaction : $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$,

The forward reaction at constant temperature is favoured by

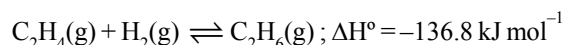
- (a) introducing an inert gas at constant volume
- (b) introducing chlorine gas at constant volume
- (c) introducing an inert gas at constant pressure
- (d) increasing the volume of the container

74. (M) Which of the following factors will favour the backward reaction ?



- (a) Addition of inert gas at constant pressure
- (b) Addition of Cl_2 gas
- (c) Increase in the temperature of reaction
- (d) Increasing the volume of the container

75. (M) For the gas phase reaction,

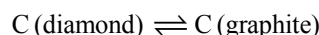


carried out in a vessel, the equilibrium concentration of C_2H_4 can be increased by :

- (a) increase in temperature (b) decrease in pressure
- (c) removing some H_2 (d) adding some C_2H_6

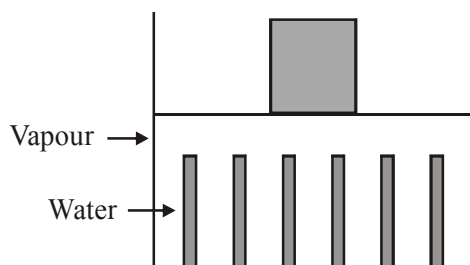
Physical Equilibria

76. (S) Densities of diamond and graphite are 3.5 and 2.3 g/mL respectively. Increase of pressure on the equilibrium



- (a) favours backward reaction
- (b) favours forward reaction
- (c) has no effect
- (d) increase the reaction rate

77. (S) Some quantity of water is contained in a container as shown in figure below. As neon is added to this system at constant pressure, the amount of liquid water in the vessel :



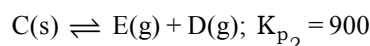
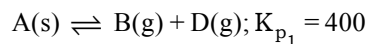
- (a) increases (b) decrease
(c) remains same (d) change unpredictably
78. (S) Solubility of a substance which dissolves with a decrease in volume and absorption of heat will be favoured by
(a) high p and high T (b) low p and low T
(c) high p and Low T (d) low p and high T
79. (S) A solution is transferred to a 1 L flask, some pure ice is added and a stopper is inserted to close the flask. After the system has reached constant temperature, there are still a few cubes of ice left in the flask, but no other solid. Which of the following statements is false relative to this system at equilibrium ?
(a) The system is a heterogeneous one.
(b) There are four phases present in the system, not including the container and stopper.
(c) The vapour pressure is constant.
(d) The temperature of the system is below 0°C.
80. (S) When a liquid in equilibrium with its vapour is heated
(a) the rate of the reaction, vapour $\rightarrow$ liquid, is increased
(b) the same equilibrium mixture is re-established
(c) a new equilibrium mixture with a higher vapour pressure is established
(d) a new equilibrium mixture is established in which the rate of evaporation is greater than the rate of condensation

Simultaneous Equilibria

81. (I) Two equilibria are simultaneously existing in a vessel at 25°C
- $$\text{NO(g)} + \text{NO}_2\text{(g)} \rightleftharpoons \text{N}_2\text{O}_3\text{(g)}; K_{p1} \text{ (say)}$$
- $$2\text{NO}_2\text{(g)} \rightleftharpoons \text{N}_2\text{O}_4\text{(g)}; K_{p2} = 8 \text{ atm}^{-1}$$

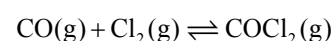
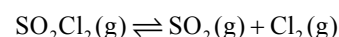
If initially only NO and NO₂ are present in a 3 : 5 mole ratio and the total pressure at equilibrium is 5.5 atm with the pressure of NO₂ is 0.5 atm, calculate K_{p1} (in 10⁻¹).

82. (I) Two solids A and C dissociate into gas products as follows.



At 25°C, the pressure over excess solid A only is 40 atm, and that over solid C only is 60 atm. Find the pressure over the solid mixture.

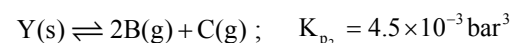
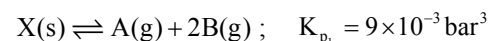
83. (S) On heating a mixture of SO₂Cl₂ and CO, two equilibria are simultaneously established :



On adding more SO₂ at equilibrium what will happen ?

- (a) Amount of CO will decrease
(b) Amount of SO₂Cl₂ and COCl₂ will increase
(c) Amount of CO will remain unaffected
(d) Amount of SO₂Cl₂ and CO will increase

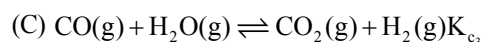
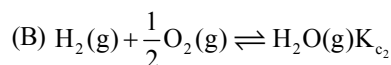
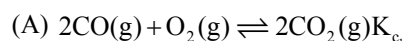
84. (S) Two solid compounds X(s) and Y(s) undergo dissociation as follows :



The total pressure of gaseous mixture at equilibrium will be :

- (a) 4.5 bar (b) 0.45 bar
(c) 0.6 bar (d) 1.2 bar

85. (S) Given the following reactions and associated equilibrium constants, select the correct expression for the third equilibrium constant in terms of the first two.



(a) $K_{c3} = \frac{K_{c1}}{K_{c2}}$ (b) $K_{c3} = \frac{(K_{c1})^{1/2}}{K_{c2}}$

(c) $K_{c3} = K_{c1} \times (K_{c2})^{1/2}$

(d) $K_{c3} = (K_{c1})^{1/2} \times (K_{c2})^{1/2}$

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

- The oxidation of SO_2 by O_2 to SO_3 is an exothermic reaction. The yield of SO_3 will be maximum if **(1981)**
 - temperature is increased and pressure is kept constant
 - temperature is reduced and pressure is increased
 - both temperature and pressure are increased
 - both temperature and pressure are reduced
- For the reaction :

$$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$$
The equilibrium constant K_p changes with **(1981)**
 - total pressure
 - catalyst
 - the amounts of H_2 and I_2 present
 - temperature
- Pure ammonia is placed in a vessel at a temperature where its dissociation constant (α) is appreciable. At equilibrium

$$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$$
 (1984)
 - K_p does not change significantly with pressure
 - α does not change with pressure
 - concentration of NH_3 does not change with pressure
 - concentration of hydrogen is less than that of nitrogen
- For the reaction ;

$$\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$$
at a given temperature, the equilibrium amount of $\text{CO}_2(\text{g})$ can be increased by **(1998)**
 - adding a suitable catalyst
 - adding an inert gas
 - decreasing the volume of the container
 - increasing the amount of $\text{CO}(\text{g})$
- For the chemical reaction

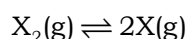
$$3\text{X}(\text{g}) + \text{Y}(\text{g}) \rightleftharpoons \text{X}_3\text{Y}(\text{g})$$
the amount of X_3Y at equilibrium is affected by **(1999)**
 - temperature and pressure
 - temperature only
 - pressure only
 - temperature, pressure and catalyst
- For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 500°C , the value of K_p is 1.44×10^{-5} when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mole litre $^{-1}$ is **(2000)**
 - $\frac{1.44 \times 10^{-5}}{(0.082 \times 500)^{-2}}$
 - $\frac{1.44 \times 10^{-5}}{(8.314 \times 773)^{-2}}$
 - $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^2}$
 - $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$
- When two reactants, A and B are mixed to give products C and D, the reaction quotient Q, at the initial stages of the reaction **(2000)**
 - is zero
 - decreases with time
 - is independent of time
 - increases with time
- At constant temperature, the equilibrium constant (K_p) for the decomposition reaction $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is expressed by $K_p = (4x^2P)(1-x^2)$, where P = pressure, x = extent of decomposition. Which one of the following statements is true ? **(2001)**
 - K_p increases with increase of P
 - K_p increases with increase of x
 - K_p increases with decrease of x
 - K_p remains constant with change in P and x
- Consider the following equilibrium in a closed container

$$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$$
At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements hold true regarding the equilibrium constant (K_p) and degree of dissociation (α) ? **(2008)**
 - Neither K_p nor α changes
 - Both K_p and α changes
 - K_p changes but α does not change
 - K_p does not change but α changes

10. $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Which is correct statement if N_2 is added at equilibrium condition ? (2006)
- (a) The equilibrium will shift to forward direction because according to IInd law of thermodynamics the entropy must increase in the direction of spontaneous reaction
- (b) The condition for equilibrium is $G_{N_2} + 3G_{H_2} = 2G_{NH_3}$ where G is Gibbs free energy per mole of the gaseous species measured at that partial pressure. The condition of equilibrium is unaffected by the use of catalyst, which increases the rate of both the forward and backward directions to the same extent
- (c) The catalyst will increase the rate of forward reaction by β
- (d) Catalyst will not alter the rate of either of the reaction.

Paragraph

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation :

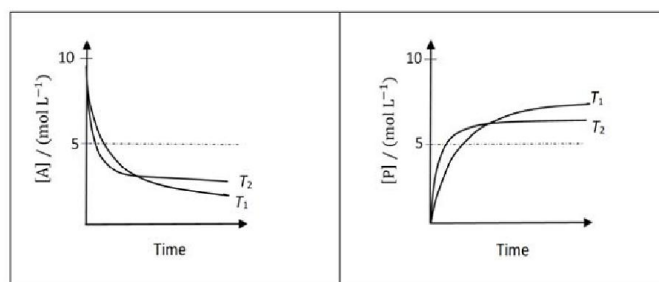


The standard reaction Gibbs energy, $\Delta_r G^\circ$, of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β . Thus, $\beta_{\text{equilibrium}}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally. (Given : $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$) (2016)

11. The equilibrium constant K_p for this reaction at 298 K, in terms of $\beta_{\text{equilibrium}}$, is
- (a) $\frac{8\beta_{\text{equilibrium}}^2}{2 - \beta_{\text{equilibrium}}}$ (b) $\frac{8\beta_{\text{equilibrium}}^2}{4 - \beta_{\text{equilibrium}}^2}$
- (c) $\frac{4\beta_{\text{equilibrium}}^2}{2 - \beta_{\text{equilibrium}}}$ (d) $\frac{4\beta_{\text{equilibrium}}^2}{4 - \beta_{\text{equilibrium}}^2}$
12. The INCORRECT statement among the following, for this reaction, is
- (a) Decrease in the total pressure will result in formation of more moles of gaseous X
- (b) At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously
- (c) $\beta_{\text{equilibrium}} = 0.7$
- (d) $K_c < 1$

Objective Questions (One or more than one correct option)

13. For the gas phase reaction
 $C_2H_4 + H_2 \rightleftharpoons C_2H_6$ ($\Delta H = -32.7 \text{ kcal}$)
carried out in a vessel, the equilibrium concentration of C_2H_4 can be increased by (1984)
- (a) Increasing the temperature
- (b) decreasing the pressure
- (c) removing some H_2
- (d) adding some C_2H_6
14. When $NaNO_3$ is heated in a closed vessel, oxygen is liberated and $NaNO_2$ is left behind. At equilibrium (1986)
- (a) addition of $NaNO_2$ favours reverse reaction
- (b) addition of $NaNO_3$ favours forward reaction
- (c) increasing temperature favours forward reaction
- (d) increasing pressure favours reverse reaction
15. The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed container and an inert gas, helium is introduced. Which of the following statements are correct ? (1989)
- (a) Concentration of SO_2 , Cl_2 and SO_2Cl_2 change
- (b) More chlorine is formed
- (c) Concentration of SO_2 is reduced
- (d) All the above are incorrect
16. For the reaction, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
The forward reaction at constant temperature is favoured by (1991)
- (a) introducing chlorine gas at constant volume
- (b) introducing an inert gas at constant pressure
- (c) increasing the volume of the container
- (d) introducing PCl_5 at constant volume
17. For a reaction, $A \rightleftharpoons P$, the plots of $[A]$ and $[P]$ with time at temperatures T_1 and T_2 are given below. (2018)



If $T_2 > T_1$, the correct statement(s) is (are)

(Assume ΔH° and ΔS° are independent of temperature and ratio of $\ln K$ at T_1 to $\ln K$ at T_2 is greater than T_2/T_1 . Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

- (a) $\Delta H^\circ < 0$, $\Delta S^\circ < 0$ (b) $\Delta G^\circ < 0$, $\Delta H^\circ > 0$
- (c) $\Delta G^\circ < 0$, $\Delta S^\circ < 0$ (d) $\Delta G^\circ < 0$, $\Delta S^\circ > 0$

Fill in the Blanks

18. For a given reversible reaction at a fixed temperature, equilibrium constant K_p and K_c are related by (1994)
19. A ten-fold increase in pressure on the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at equilibrium, results in in K_p . (1996)
20. For a gaseous reaction $2B \longrightarrow A$, the equilibrium constant K_p is to/than K_c .

True/False

21. When a liquid and its vapour are at equilibrium and the pressure is suddenly decreased, cooling occurs. (1984)
22. If equilibrium constant for the reaction, $A_2 + B_2 \rightleftharpoons 2AB$, is K , then for the backward reaction $AB \rightleftharpoons \frac{1}{2}A_2 + \frac{1}{2}B_2$, the equilibrium constant is $\frac{1}{K}$. (1984)
23. Catalyst makes a reaction more exothermic. (1987)
24. The rate of an exothermic reaction increases with increasing temperature. (1993)

Subjective Questions

25. One mole of nitrogen is mixed with three moles of hydrogen in a four litre container. If 0.25 per cent of nitrogen is converted to ammonia by the following reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$
then calculate the equilibrium constant, K_c in concentration units. What will be the value of K_c for the following equilibrium?

$$\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightleftharpoons NH_3(g) \quad (1981)$$
26. One mole of Cl_2 and 3 moles of PCl_5 are placed in a 100 litre vessel heated at $227^\circ C$. The equilibrium pressure is 2.05 atmosphere. Assuming ideal behaviour, calculate the degree of dissociation for PCl_5 and K_p for the reaction

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \quad (1984)$$
27. The equilibrium constant of the reaction

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$
at $100^\circ C$ is 50. If a one litre flask containing one mole of A_2 is connected to a two litre flask containing two moles of B_2 , how many mole of AB will be formed at $373^\circ C$? (1985)
28. At a certain temperature, equilibrium constant (K_c) is 16 for the reaction;

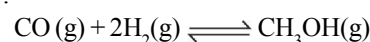
$$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$$
If we take one mole of each of all the four gases in a one litre container, what would be the equilibrium concentration of NO and NO_2 ? (1987)
29. N_2O_4 is 25% dissociated at $37^\circ C$ and one atmosphere pressure. Calculate (i) K_p and (ii) the percentage dissociation at 0.1 atmosphere and $37^\circ C$. (1988)

30. The equilibrium constant K_p of the reaction,

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$
is 900 atm at 800 K. A mixture containing SO_3 and O_2 having initial pressure of 1 and 2 atm respectively is heated at constant volume to equilibrate. Calculate the partial pressure of each gas at 800 K. (1989)

31. For the reaction

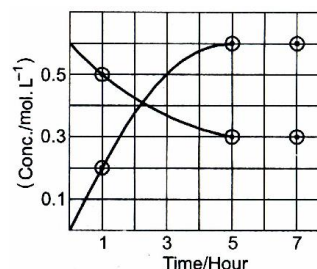
$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$
hydrogen gas is introduced into a five litre flask at $327^\circ C$, containing 0.2 mole of $CO(g)$ and a catalyst, until the pressure is 4.92 atm. At this point 0.1 mole of $CH_3OH(g)$ is formed. Calculate the equilibrium constant, K_p and K_c . (1990)
32. 0.15 mole of CO taken in a 2.5 L flask is maintained at 750 K along with a catalyst so that the following reaction can take place:



Hydrogen is introduced until the total pressure of the system is 8.5 atm at equilibrium and 0.08 mole of methanol is formed. Calculate (i) K_p and K_c and (ii) the final pressure if the same amount of CO and H_2 as before are used, but with no catalyst so that the reaction does not take place. (1993)

33. The progress of reaction,

$$A \rightleftharpoons nB$$
with time, is represented in fig. given below. Determine



- (i) the value of n
(ii) the equilibrium constant, K and
(iii) the initial rate of conversion of A . (1994)
34. The degree of dissociation is 0.4 at 400 K and 1.0 atm for the gaseous reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$. Assuming ideal behaviour of all the gases, calculate the density of equilibrium mixture at 400 K and 1.0 atm (relative atomic mass of $P = 31.0$ and $Cl = 35.5$) (1998)
35. When 3.06 g of solid NH_4SH is introduced into a two litre evacuated flask at $27^\circ C$, 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide.
(i) Calculate K_c and K_p for the reaction at $27^\circ C$.
(ii) What would happen to the equilibrium when more solid NH_4SH is introduced into the flask? (1999)

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

1. (c)	2. (b)	3. (d)	4. (d)	5. (d)	6. (c)	7. (a)	8. (a)	9. (c)	10. (d)
11. (d)	12. (b)	13. (b)	14. (d)	15. (a)	16. (b)	17. (b)	18. (a)	19. (c)	20. (c)
21. (d)	22. (c)	23. (c)	24. (b)	25. (d)	26. (a)	27. (d)	28. (c)	29. (c)	30. (a)
31. (d)	32. (a)	33. (c)	34. (b)	35. (c)	36. (a)	37. (d)	38. (b)	39. (d)	40. (b)
41. (c)	42. (a)	43. (b)	44. (a)	45. (d)	46. (a)	47. (b)	48. (d)	49. (a)	50. (a)
51. (a)	52. (a)	53. (d)	54. (d)	55. (b)	56. (a)	57. (c)	58. (d)	59. (b)	60. (d)
61. (b)	62. (b)	63. (d)	64. (b)	65. (d)	66. (d)	67. (d)	68. (a)	69. (d)	70. (a)
71. (c)	72. (c)	73. (c)	74. (a)	75. (c)	76. (b)	77. (c)	78. (d)	79. (d)	80. (d)
81. (d)	82. (c)	83. (d)	84. (c)						

EXERCISE - 2 : (Previous year JEE Mains Questions)

1. (a)	2. (c)	3. (d)	4. (d)	5. (a)	6. (b)	7. (a)	8. (d)	9. (d)	10. (d)
11. (a)	12. (a)	13. (a)	14. (c)	15. (a)	16. (a)	17. (d)	18. (b)	19. (c)	

JEE Mains Online

1. (d)	2. (c)	3. (a)	4. (d)	5. (a)	6. (c)	7. (d)	8. (a)	9. (b)	10. (d)
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EXERCISE - 3 : (Advanced Objective Questions)

1. (a)	2. (a)	3. (b,c)	4. (c)	5. (b)	6. (a)	7. (a)	8. (a)	9. (a,c)	11. (c)
12. (a)	13. (a)	14. (0004)	15. (a)	16. (c)	17. (b)	18. (b)	19. (c)	20. (c)	21. (a)
22. (a)	23. (b)	24. (A-s), (B-p), (C-q), (D-r)			25. (b)	26. (b)	27. (b)	28. (a,c,d)	
29. (a,b)	30. (a)	31. (a,d)	32. (d)	33. (c)	34. (c)	35. (a)	36. (a)		

37. (A-p,q,s), (B-q,r), (C-q), (D-p,s) 38. (0008) 39. (0008) 40. (0006) 41. (b) 42. (c) 43. (b) 44. (d)
45. (d) 46. (b) 47. (d) 48. (a,c) 49. (a) 50. (b) 51. (b) 52. (b) 53. (a) 54. (a)
55. (d) 56. (a,c) 57. (a,b,d) 58. (c,d) 59. (b,d) 60. (b,c,d) 61. (a,b,c) 62. (a,c) 63. (a,c) 64. (c)
65. (c) 66. (d) 67. (a) 69. (A-p,r,s), (B-q,r), (C-r), (D-r) 70. (d) 71. (c) 72. (b)
73. (c,d) 74. (a,c,d) 75. (a,b,c,d) 76. (a) 77. (b) 78. (a) 79. (b) 80. (c) 81. (0004)
82. (0072) 83. (d) 84. (b) 85. (b)

EXERCISE - 4 : (Previous year JEE Advanced Questions)

1. (b) 2. (d) 3. (a) 4. (d) 5. (a) 6. (d) 7. (d) 8. (d) 9. (d) 10. (b)
11. (b) 12. (c) 13. (a,b,c,d) 14. (c,d) 15. (d) 16. (bcd) 17. (a,c) 18. $K_p = K_c (RT)^{\Delta n}$
19. no change 20. smaller 21. (T) 22. (F) 23. (F) 24. (F) 25. $0.468 \text{ L}^2 \text{ mol}^{-2}$, 0.68
26. 0.33 27. 1.76 29. 0.266 atm; 63% 30. $\frac{175}{87} \text{ atm}$, $\frac{85}{87} \text{ atm}$ 31. $K_p = 0.11 \text{ atm}^{-2}$, $K_c = 277.77 \text{ M}^{-2}$
32. 220 33. (i) $n = 2$, (ii) 1.2 34. 4.54g/L 35. 8.1×10^{-5} , 4.84×10^{-2} , (ii) No effect

Dream on !!

